

Simulating link-level interrupted flow traffic dynamics and the comparison between different models for urban road networks

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ABSTRACT

Dynamic traffic models that reproduce the pattern of congestion propagation in urban road networks are essential for the evaluation of traffic management strategies and prediction of traffic states. Many approaches developed in the past simplified the movement of vehicles and the propagation of traffic waves. The overlooked problems can be particularly influential for the simulation of interrupted flow with traffic signal control due to the frequent accumulation and dissipation of the waiting queue on a road link. This study first reviews state-of-the-art macroscopic and mesoscopic link-level urban traffic flow models and provides a comparative discussion of their similarities, differences, and gaps. We then put forward an event-based mesoscopic model to simulate link-level interrupted flow traffic dynamics (LIFT). The model (i) simulates the transmission of vehicles between links based on the demand and supply of exit and entry events, (ii) monitors queue spillback through the consideration of backward traveling spaces, and (iii) adheres to first-in-first-out at intersections for congested situations. Taking the outcomes generated from microscopic simulation as ground-truth, the case studies show that LIFT outperforms the other models by accurately capturing the evolution of link densities and mean path speeds in congested conditions. It is reliable even in complex scenarios with diverge blocking phenomena and desired speed heterogeneity. Without having to simulate the interaction between individual vehicles, the model also becomes much more computationally efficient than microscopic simulation. It can be applied in simulation-based optimization or control problems which require the consideration of finer-level details that macroscopic models are unable to offer.

1. Introduction

The spatiotemporal propagation pattern of traffic congestion in an urban road network has been analyzed with various macroscopic models that contain plenty of simplification and microscopic simulation tools that require intensive computation for relatively large networks. Fig. 1 summarizes traffic flow modeling approaches at different levels of aggregation and their principles.

Region-level urban traffic modeling based on network fundamental diagrams (NFDs) has become a popular approach in recent years considering its superior computation efficiency for network-wide traffic management strategies, including perimeter control [1] and toll pricing [2]. Johari et al. [3] provided an in-depth review on the development of this type of approach. Nevertheless, the shape of an NFD is subject to changes of the origin–destination (OD) demand and trip pattern within the network [4–6], indicating that it may not be a suitable option to be used in traffic control strategies when adaptive route choice behavior is considered. Moreover, changes in road infrastructure supply, e.g., road closure or dynamic bus lane allocation, also affect the shape of an NFD.

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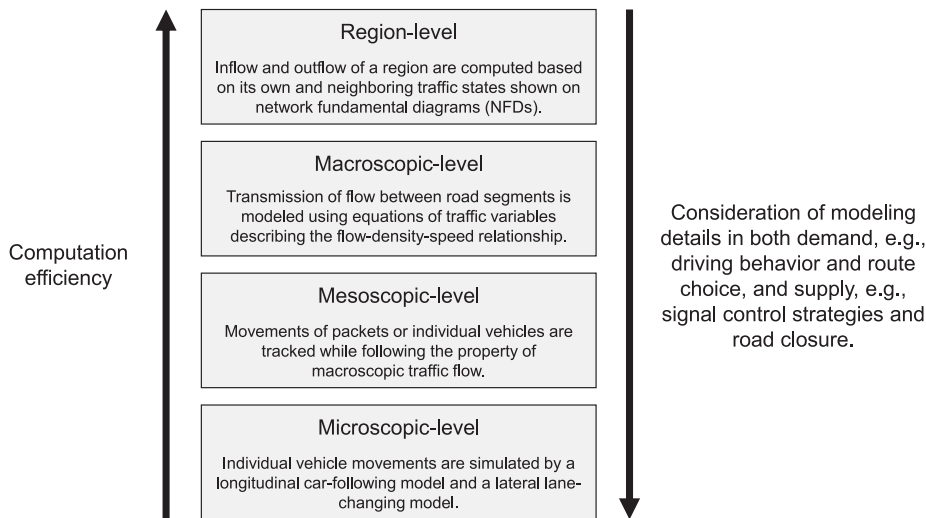


Fig. 1. Classification of traffic flow models based on levels-of-aggregation.

Defining regions with sufficiently homogeneous traffic dynamics so that the NFD can be representative is also a major obstacle when attempting to use these approaches [7]. In view of these limitations, approaches that simulate urban traffic dynamics at a finer level are still of significant importance. They play a founding role in the development of region-level traffic models as well.

At a finer level relative to region-level traffic flow models, congestion propagation can be monitored by investigating the evolution of traffic density on each link in the urban road network. The average link travel speed decreases as the density increases after a critical point, as reflected on the trapezoidal fundamental diagram (FD) of urban traffic flow [8]. When a road link reaches the saturated state, link travel time increases drastically. Once the link becomes oversaturated, spillback occurs due to the long vehicle queue and hence reduces the flow that can enter from the upstream. The congestion then begins from a single link and spreads throughout the network. The linkage of congested clusters eventually leads to the breakdown of network traffic flow. Following such a concept, Saberi et al. [9] described the process of congestion propagation and dissipation dynamics with a system of ordinary differential equations, which was inspired by the spread of contagious diseases within a population. Ambühl et al. [10] also modeled this process and concluded that the collapse of network traffic flow occurs when the percolation of congested links transforms into a large connected cluster.

An accurate and efficient traffic model that can reproduce the congestion dynamics described above and help identify bottleneck locations is useful for the design of a road network or the development of urban traffic management strategies. Mesoscopic traffic simulation models that follow the property of macroscopic traffic flow and in the meantime possess the merit of microscopic simulation have gained a lot of attention among modelers. However, the performance of these mesoscopic models with regard to accuracy has not been thoroughly discussed before.

In this study, a mesoscopic traffic flow model is proposed to simulate link-level interrupted flow traffic dynamics (LIFT) exhibited by urban road links with traffic signal control. The model, hereinafter referred to as LIFT, is anticipated to outperform other macroscopic and mesoscopic models in capturing the evolution of link traffic states before, during, and after congestion. The following are the main features of LIFT.

- Simulating vehicle movements between links efficiently based on the demand and supply of entry and exit events at boundaries
- Monitoring the occurrence of queue spillback accurately by utilizing the concept of backward traveling spaces
- Adhering to exact first-in-first-out (FIFO) at urban intersections particularly for diverge blocking situations caused by queue spillback

The remainder of the paper is structured as follows. In Section 2, we first review past studies on macroscopic traffic modeling and mesoscopic traffic simulation and point out the problems which still require careful attention for urban traffic flow modeling in particular. Section 3 then explains the model formulation of LIFT, and Section 4 describes the design of case studies. The comparison between all the selected models is then presented in Section 5, while Section 6 summarizes the contribution of this study.

2. State-of-the-art models

In this section, we review several macroscopic and mesoscopic traffic models and their applications in the context of urban traffic flow. The drawbacks when using these models, especially magnified by the distinct property of interrupted traffic flow dynamics, are highlighted.

2.1. Macroscopic-level urban traffic flow modeling

Macroscopic traffic flow models describe traffic dynamics by capturing the movement of flow (aggregated vehicle behavior). Many studies related to urban traffic modeling adapted macroscopic models that were originally proposed for uninterrupted flow on motorways. For instance, Simoni and Claudel [11] modified the Hamilton–Jacobi formulation of the Lighthill–Whitham–Richards (LWR) model, which was originally proposed in [12], to evaluate the effect of traffic signal control. There were also studies that extended the cell transmission model (CTM) [13], a spatiotemporal discretization of the LWR model solved with the Godunov scheme [14], for urban traffic flow [15–17]. The delay at signalized intersections and the reduced queue discharge rate caused by the switching of signal phases are considered. However, deliberately capturing the progression of uninterrupted vehicle flow between cells on urban links has little positive contribution to the modeling accuracy and is considered unnecessarily refined. The computation requirement also becomes a concern when applying CTM to a large-scale network due to the division of a road link into multiple cells.

Compared to CTM, link transmission model (LTM), which is proposed based on Newell’s formulation of the LWR model [18], has gained a lot of attention in dynamic traffic modeling due to its improved accuracy derived from continuous-space consideration within the link and better computation efficiency by taking the entire road link as a unit [19]. In the extended LTM formulation for urban traffic flow proposed in [20], intersection delay is considered by quantifying the time-aggregated impact of signals on link outflow capacity. The outflow capacity is reduced based on the green-to-cycle ratio. This type of approach was later called continuous signalized node modeling, which is a simplified yet efficient way to model traffic flow in urban road networks [21–23], especially for cases with complicated intersection geometries (multiple incoming and outgoing links) and signal timing optimization or control purposes using LTM [24,25]. However, as this study only focuses on the discussion of modeling accuracy, only the precise “on-and-off” signal timing model would be used.

In macroscopic CTM and LTM, the time step size and the cell/link length must follow the Courant–Friedrichs–Lewy (CFL) condition. For traffic modeling of an urban road network with various link lengths, it is quite impractical for the model to comply with this requirement, especially when those short links act as critical bottlenecks in the network and should not be combined into a long road link. In addition, to model the on-and-off signal control, traffic signal timing imposes another specific requirement on these models that the time step size should be a common divisor of all the signal phases. This either limits the possible signal timing plans that the model can consider or leads to a rather small time step size and again causes high computational cost. Therefore, macroscopic LTM has rarely been applied to large-scale networks with complex signal timing plans and road geometries. To overcome this kind of limitation, methods that solve LTM in an event-based resolution [26] or using an iterative algorithm with linear interpolation on a solution grid [27] can be considered.

For the modeling of interrupted traffic flow in particular, Gazis and Potts [28] proposed the store-and-forward (SaF) model, where each road link is considered a queue system of vehicles. The signal timing plan is explicitly included as parameters in the discrete-time model formulation. Therefore, the model has often been utilized to optimize traffic signal timing [29–31]. However, it is known that the simplified assumption of a vertical link queue ignores the time required for vehicles to travel through the link, which deteriorates the model performance in under-saturated conditions.

To consider link travel time and available storage space in the downstream link, Lin et al. [32] proposed the *S* model by separating the moving and queuing vehicles. The transfer of vehicle flows within a link, hence, also needs to be modeled at every time step. By keeping track of the queue tail position, they aimed to capture the stop-and-go movement in detail. There were also other studies that utilized link-level macroscopic urban traffic models for other problems. Tsitsokas et al. [33] extended the SaF model and integrated it into an optimization problem for dedicated bus lane allocation. Different from the original SaF model, the extended SaF (eSaF) additionally incorporates the computation of delay within the link by separating the moving and queuing vehicles, the same feature applied in the *BLX* model and *S* model [32]. Therefore, the transfer of flows between these two groups within a link needs to be updated at every time step. Wei et al. [34] proposed a link-based flow model that follows a similar principle but consider time-varying free-flow speed to better capture its impact on queue outflow. However, for these models, the transmission between moving and queuing vehicles can be problematic due to the discrete-time setup. Moreover, these models neglect the fact that the number of waiting vehicles does not always represent the queue tail position. During the dissipation of a queue, the number of waiting vehicles decreases, while the queue tail position remains. Hence, the length of the upstream free-flow segment on the link should stay the same in such a situation. The problem can be understood by looking at the two cases in the exemplary Fig. 2 where there is the same number of vehicles in the downstream queue but different lengths of the available space (or free-flow segment) at the upstream of the link.

In Fig. 2a, a static queue formed during the red phase of the downstream traffic signal is shown. In eSaF, the available space for entry vehicles coming from the upstream links is estimated by computing the difference between the maximum number of vehicles that the link can accommodate and the actual number on the link. However, when the waiting vehicles in the queue are discharging, the available space remains unchanged even though the number of vehicles in the queue decreases at the beginning of the green phase, as illustrated in Fig. 2b. Many macroscopic traffic models developed in the past assume a theoretical stationary traffic state on a link and presume that the available space length can be directly derived by simply knowing the number of vehicles in the queue. Without considering the dynamics at the downstream end of the link in a simulation model, the overestimation of available space would falsely speed up the transmission of vehicles between links in oversaturated conditions.

Besides the constraints and problems mentioned so far, it is also known that macroscopic traffic models face other difficulties, while notable ones being the FIFO adherence at the vehicle-level and the inclusion of driving behavioral heterogeneity among vehicles in the multi-commodity setup. In order to achieve multi-commodity flow modeling, Raadsen and Bliemer [35] proposed the

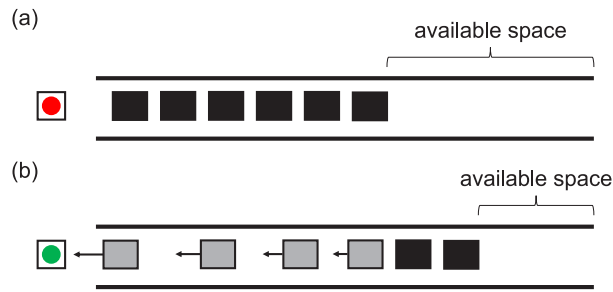


Fig. 2. Overestimation of available space in the eSaF model.

concept of mixture propagation and distribution in a continuous-time event-based LTM. However, there was no discussion pertaining to FIFO adherence in the study. The violation of FIFO may become significant when there is queue spillback in the network. To ensure perfect FIFO or minimize FIFO violation for macroscopic traffic modeling, interested readers are referred to Bar-Gera and Carey [36,37]. The in-depth discussion of this aspect is considered out-of-the-scope here.

Although FIFO may not always hold when overtaking is considered in multi-lane conditions, the interrupted flow dynamics in an urban road network reduces overtaking frequency due to the division of links based on signalized intersections (nodes). Compared to the case of uninterrupted traffic flow on motorways, vehicles cannot easily perform lane-changing to gain speed. Most of the lane-changing behaviors in an urban road network are actually strategic lane-changing for downstream turning directions. Therefore, FIFO becomes important to follow when modeling interrupted traffic flow. In addition, FIFO violation is of particular concern for traffic flow modeling of a limited urban road environment in which there are no dedicated lanes to accommodate different turning directions. When vehicles in different commodities appear on a single-lane road, macroscopic models are unable to maintain the order of commodities exiting the link, making it difficult to precisely simulate the rapid propagation of congestion due to queue spillback in this type of urban road network.

2.2. Mesoscopic-level urban traffic flow simulation

The previous subsection mainly raises the concern related to the ignorance of thorough link-level queue dynamics when using several macroscopic traffic models. Such a problem can also exist in mesoscopic traffic models.

Based on the definition of mesoscopic traffic simulation in this study, individual vehicles or packets, i.e., groups of vehicles, are tracked while their movements follow the property of macroscopic traffic flow instead of behavioral description in microscopic models. Compared to macroscopic models, these models possess the advantage of exact FIFO in multi-commodity flow modeling with route choice and the ability to consider vehicle acceleration dynamics and behavioral heterogeneity at a finer level.

Di Gangi [38] performed dynamic network loading by modeling traffic flow as continuous packets, in which vehicles are uniformly distributed in both time and space. To avoid numerical difficulties when link density changes between time steps, more studies adopted the discrete packet approach, which puts multiple vehicles in a packet at one point in time [39,40]. Mesoscopic simulation tools, such as DYNASMART [41] and DynaMIT [42], can be regarded as implementations of these discrete packet approaches with the individual vehicle representation.

In TRAFFMED [40] and DynaMIT, an arc is divided into a running segment and a queuing segment, as illustrated in Fig. 3, while the queue tail location is updated at each time step based on the loading condition. However, the problem of overestimated available space, identical to the problem in the eSaF mentioned in the previous subsection, again arises. These models also ignore the fact that the number of vehicles in the queueing segment cannot be directly translated into the queue tail location. The partition of an urban road link into free-flow (running) and congested (queueing) segments is a problematic approach.

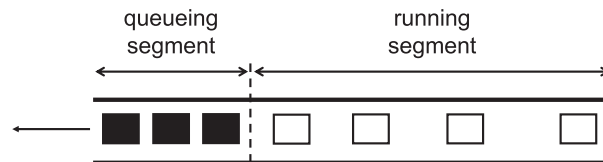


Fig. 3. Representation of the queueing model in TRAFFMED and DynaMIT.

Moreover, in these models, the speed of vehicles on the running segment is determined in an aggregated manner based on a function of density following the assumption of homogeneous traffic state on the segment. In reality, this assumption cannot hold on a road link due to the unsteadiness of its traffic state, similar to the hysteresis formed on an FD [43]. Especially on a signalized urban road, such an effect is amplified by the frequent stop-and-go movements caused by traffic signal control. Determining the link speed based on a speed-density function problematically slows down vehicles and results in very low outflow demand from the running segment to the queueing segment. Therefore, in the replication of DynaMIT for the case study in this work, the speed

function is replaced with the free-flow speed to prevent such a problem. We also found that the description of the original DynaMIT in [44] complies better with this logic.

In Simulation of Urban MObility (SUMO), an open-source package [45], the mesoscopic simulation function does not rely on any car-following model but controls the exit headway at boundaries according to the states of the current and downstream edges [46], as shown in Fig. 4. Vehicles are ready to exit the link after the free-flow link travel time. Each edge is classified as free-flow or congested based on its density (occupancy). The update of headway is carried out based on the edge states at every discrete time step. Between two congested edges, the entry headway is computed as the time required for an empty space to travel backward to the upstream end. Unlike all the aforementioned mesoscopic models, this kind of approach does not require tracking of the exact individual vehicle or packet location on the link to determine the link traffic state and speed, ensuring faster computation.

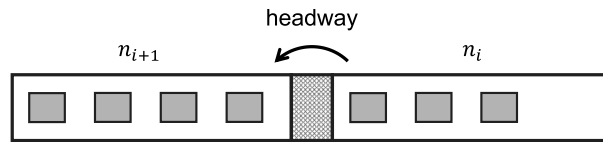


Fig. 4. Representation of Eissfeldt's model.

Nevertheless, the model seeks to consider the edge-internal heterogeneity, which is a feature of LTM, but in the meantime contradictorily assumes traffic homogeneity within an edge so that the edge state can be known by simply comparing its density with the critical density, which is a concept applied in CTM. The same number of vehicles on a short edge can represent multiple conditions, e.g., vehicles are equally spaced, the queue is accumulating, or the queue is dissipating. Since it is impossible to tell the vehicle distribution within an edge in a mesoscopic model, spillback condition is not explicitly monitored in the discrete-time formulation of the model.

Consequently, Eissfeldt's model contains several downsides. First, when the headway is determined based on the travel time of a backward space for "congested" links instead of "fully-jammed" links, the delay is overestimated. Second, while the number of vehicles on the link decreases because of the exit of a vehicle, the entry headway changes regardless of the headway computed in the previous time step. Vehicles may enter the link even though the empty space is still propagating backward and has not yet arrived at the upstream end. The memoryless setting in this aspect leads to a lack of spatial-temporal correlation when determining the headway. As a result, spillback queues dissipate earlier than in reality, and links accommodate more vehicles than the number that its available space can store. The influence of such problems could be significant particularly for urban traffic flow because of the frequent queue accumulation and dissipation in congested conditions.

Charypar et al. [47] proposed a queue-based model with a similar concept in a discrete-event formulation. The difference compared to the model proposed in [46] is that the entry headway of a congested link does not depend on the link density anymore but the exact time when a space is available at the upstream boundary of the link. It can be regarded as the mesoscopic implementation of the two-queue LTM concept in an event-based formulation, which was proposed in [48] and described in [49] in detail. The model was originally included in MATSim [50] as the function Java Discrete Event Queue Simulation (JDEQSIM), but it does not function in scenarios with traffic lights. Also, JDEQSIM has not been widely-adopted and may not be supported anymore due to lack of maintenance. Agarwal et al. [51] further utilized its discrete-time counterpart and extended it to consider mixed traffic conditions. Still, the validity of this approach was never tested. Knowing that time-discretization only yields an approximation of the exact solution to the LWR model and sacrifices accuracy, de Souza et al. [52] later proposed another form of mesoscopic LTM in a discrete-time formulation by associating the cumulative inflow and outflow to each vehicle so that their movements can be handled individually and in the meantime ensuring the consistency between the continuous cumulative curves in macroscopic LTM and the modeled discrete cumulative curves because the flow between boundaries should always be an integer. This approach can be more computationally efficient than the event-based formulation.

Most studies on mesoscopic traffic flow modeling mainly discussed its contribution to computation efficiency for large-scale network case study. The validation highly relied on calibrating the route choice behavior and OD demand to fit the aggregated empirical measurements, while the verification of its simulated traffic dynamics was often neglected. As a result, the problem of oversimplifying traffic flow dynamics was covered up.

It should be noted that commercial traffic simulation tools, including Dynameq [53], PTV Vissim [54], and Aimsun [55], provide event-based mesoscopic traffic simulation functions. However, longitudinal behaviors in these software applications are based on a car-following model with relatively few parameters [56]. Therefore, following the definition in this study, they are regarded as microscopic models, albeit much simplified.

2.3. Summary

According to our review of traffic flow models for urban road networks, the concept of LTM is considered the most suitable modeling principle for urban traffic flow due to its ability to capture link-level interrupted flow traffic dynamics at both macroscopic and mesoscopic levels. In this study, we first implement a mesoscopic traffic flow model by building on the approach developed in [47]. Instead of only the software architecture introduced in that paper, a clear description of the model is provided in Section 3.

Several other link-level macroscopic and mesoscopic models, including SaF, eSaF, LTM, DynaMIT, and Eissfeldt's model (mesoscopic simulation function in SUMO at the moment of this study), are selected for comparison. The accuracy of all these traffic flow models, which has seldom been discussed in research endeavors with similar purposes in the past, is evaluated through case studies and compared with ground-truth outcomes generated using the microscopic simulation function in SUMO.

3. Model description

In this section, the model formulation of LIFT is introduced. Table 1 summarizes the notations used to describe the traffic dynamics on a single link i in this paper. Detailed definitions and calculations are described in the corresponding subsections.

Table 1
List of notations.

Parameters			
L_i	Link length	V^f	Free-flow speed
q^{sat}	Saturation flow rate	V^{space}	Backward space traveling speed
l^{eff}	Effective vehicle length	r	Reaction time
k_i^{max}	Maximum (jammed) number of vehicles	G_i	Green length
C_i	Cycle length		
Variables			
$t_{N_i^{\text{in}}-1}^{\text{entry}}$	Recorded entry time of the previous vehicle	$t_{N_i^{\text{out}}-1}^{\text{exit}}$	Recorded exit time of the previous vehicle
$s_{N_i^{\text{in}}}^{\text{entry}}$	Entry supply time	$s_{N_i^{\text{out}}}^{\text{exit}}$	Exit supply time
$d_{N_i^{\text{out}}}^{\text{exit}}$	Exit demand time	k_i^{veh}	Number of vehicles on the link
k_i^{space}	Number of spaces on the link	k_i^{sb}	Threshold number indicating spillback
$p_{N_i^{\text{out}}}$	Position of the next exiting vehicle	$p_{S_i^{\text{next}}}$	Position of the next arriving backward space
ϕ_i	Time-in-cycle	t	Current simulation time
Objects			
N_i^{in}	Order of the next entering vehicle	N_i^{out}	Order of the next exiting vehicle
S_i^{next}	Order of the next arriving backward space		

3.1. General concept

Instead of tracking the exact position of each individual vehicle, LIFT considers each road link as a queue system that stores vehicles and updates the remaining link travel distance of each vehicle at every event step. Fig. 5 illustrates the modeling principle on a single link i . Unlike in microscopic models, vehicles simulated in LIFT do not affect others movement. Instead of their exact position, only the link where they are located is tracked. Vehicles are ready to exit the link after spending the minimum link travel time, which can be calculated based on the link length L_i and the free-flow speed V^f (speed limit or desired speed). Under such a formulation, FIFO can be fully ensured as the order of vehicles in the queue is maintained. N_i^{in} and N_i^{out} are the orders of the vehicles that are about to enter and exit the link, respectively.

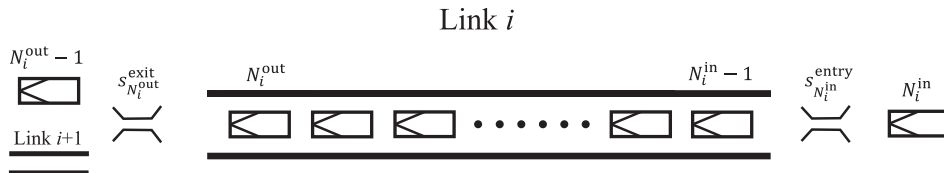


Fig. 5. Representation of LIFT on a single link.

At the upstream boundary of the link, the entry is limited by an entry supply time $s_{N_i^{\text{in}}}^{\text{entry}}$, which can normally be determined based on the saturation flow rate q^{sat} . A vehicle always maintains a minimum headway of $\frac{1}{q^{\text{sat}}}$ with the previous entry vehicle $N_i^{\text{in}} - 1$. If there is a queue spillback condition, meaning that there is an idle vehicle at the link tail blocking the entry vehicles, the entry supply time becomes the time when the next backward traveling space reaches the upstream link boundary.

3.2. Backward traveling space

It is important to note that spillback occurs not only when the link is fully occupied, indicating that the maximum number of vehicles on the link is reached, as illustrated in Fig. 6a. In Fig. 6b, a spillback condition is shown with fewer vehicles than the maximum number of the link during queue dissipation. The misconsideration of a dissipating queue leads to an overestimated available space for entry vehicles at the upstream boundary of a link. For motorway traffic flow, such dissipation does not often occur, while it constantly happens for urban traffic flow between signal phase transitions and should therefore be even more carefully considered.

Considering the aforementioned phenomenon, a method to monitor the spillback condition on an urban link can be designed by using not only the number of vehicles but also the number of backward traveling spaces on the link. The concept of backward traveling space indicates that a space is created at the downstream link boundary due to the exit movement of a vehicle. The space

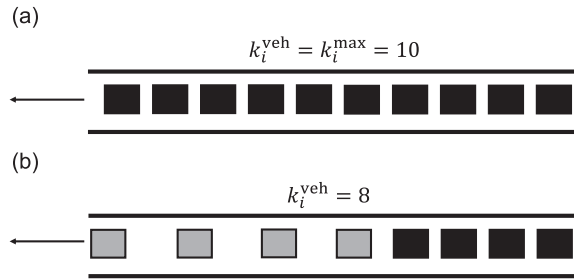


Fig. 6. Visualization of a full link and the waiting queue dissipation on a link.

then travels upstream until it arrives at the upstream end of the link. Similarly to the movement of vehicles described in Section 3.1, the spaces would not interact with each other while propagating along the link. What matters is the moment at which they reach the upstream end of the link as it is when a new space at the boundary becomes available for the next entering vehicle. The speed of the space can be determined by $V^{\text{space}} = l^{\text{eff}}/r$, where l^{eff} denotes the effective length of a vehicle, i.e., the sum of the vehicle length and safety gap, and r is the reaction time for the vehicle to start moving after the preceding vehicle. Note that the value of r should theoretically result in a V^{space} that matches the backward wave speed shown on the triangular FD.

As illustrated in Fig. 7, the void spaces between the discharging vehicles are created when a virtual space passes through the vehicle. A virtual space has the size of the effective vehicle length, including the vehicle length and a standstill distance gap. That is, for a single-lane link, the link length consists of the total effective length of all vehicles and virtual spaces on the link. Therefore, the threshold number of vehicles determining whether there is queue spillback on the link is a dynamic variable and can be calculated by $k_i^{\text{sb}} = k_i^{\text{max}} - k_i^{\text{space}}$. If the number of vehicles on the link k_i^{veh} is equal to the threshold, this indicates that there is a spillback condition. No vehicles are allowed to enter the link under such a circumstance until the next backward traveling space arrives at the upstream boundary. Therefore, the remaining travel distance for every space is tracked at every event step.

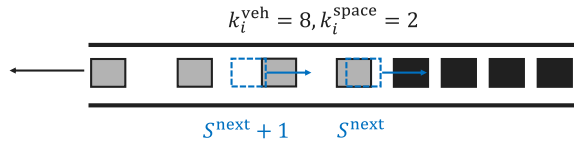


Fig. 7. Tracking of the backward traveling spaces (blue dashed).

LIFT simulates the propagation of backward traveling spaces to monitor the available space at the upstream end of a link and update the entry supply time, which would be introduced in the next subsection. This approach can also be understood as the mesoscopic counterpart of the calculation of the state variable “vacancy size”, which indicates the occurrence of queue spillback on a link, mentioned in [57].

3.3. Update of supply and demand for entry and exit at link boundaries

Accordingly, the entry supply time (earliest possible time) for the next entry vehicle N_i^{in} on link i can be written as Eq. (1). $t_{N_i^{\text{in}-1}}^{\text{entry}}$ denotes the actual entry time of the previous entry vehicle. When there is a spillback condition, the model takes into account the arrival time of the next backward traveling space S_i^{next} , as calculated in the second case in Eq. (1). Note that t indicates the current time in the model, while $p_{S_i^{\text{next}}}$ represents the position of the next arrival space.

$$s_{N_i^{\text{in}}}^{\text{entry}} = \begin{cases} t_{N_i^{\text{in}-1}}^{\text{entry}} + \frac{1}{q^{\text{sat}}}, & \text{if } k_i^{\text{veh}} < k_i^{\text{sb}} \\ \max(t_{N_i^{\text{in}-1}}^{\text{entry}} + \frac{1}{q^{\text{sat}}}, t + (L_i - p_{S_i^{\text{next}}})/V^{\text{space}}), & \text{otherwise} \end{cases} \quad (1)$$

For the exit at the downstream end of the link, the time of the exit demand can be calculated by the remaining travel distance of the next exit vehicle N_i^{out} , as written in the first case in Eq. (2). An exit request is created when the next exit vehicle reaches the boundary. In addition, the vehicle may need to wait until the start of the next green phase if the downstream signal is still red. Hence, the remaining time for the signal to turn green is considered. In Eq. (2), $p_{N_i^{\text{out}}}$ denotes the remaining distance for vehicle N_i^{out} to travel on link i , and G_i is the green length in the signal timing plan at the downstream of link i , while ϕ_i represents the time-in-cycle of the signal. $\phi_i = 0$ indicates the start of the green phase at the downstream of link i , while C_i being the cycle length of the signal timing plan. During red, the next exiting vehicle must wait until the signal turns green at $t + (C_i - \phi_i)$ to exit the link. Depending on the elements in the signal timing plan, including cycle length, green length, and offset, ϕ is different for every link

in the same event step.

$$d_{N_i^{\text{out}}}^{\text{exit}} = \begin{cases} t + (L_i - p_{N_i^{\text{out}}})/V^f, & \text{if } \phi_i \leq G_i \\ \max(t + (L_i - p_{N_i^{\text{out}}})/V^f, t + (C_i - \phi_i)), & \text{otherwise} \end{cases} \quad (2)$$

The actual exit time is limited by the exit supply, which is normally influenced by the saturation flow rate. Vehicles N_i^{out} can only exit $\frac{1}{q_{\text{sat}}}$ after the actual exit time of the previous exit vehicle at $t_{N_{i-1}^{\text{out}}}^{\text{exit}}$. In addition, the entry supply of the downstream link for vehicle N_i^{out} , which is denoted by $i + 1$ in Eq. (3) also needs to be checked. If there is a spillback condition on link $i + 1$, the downstream link of the next exit vehicle, the vehicle would need to wait until the downstream waiting queue is resolved. Vehicles behind N_i^{out} are, therefore, blocked due to the FIFO principle.

$$s_{N_i^{\text{out}}}^{\text{exit}} = \max(t_{N_{i-1}^{\text{out}}}^{\text{exit}} + \frac{1}{q_{\text{sat}}}, s_{N_{i+1}^{\text{in}}}^{\text{entry}}) \quad (3)$$

The actual time of the next exiting vehicle N_i^{out} can then be determined by comparing the updated demand and supply times, as shown in Eq. (4). The same logic also applies to the determination of the actual time of the next vehicle entry event.

$$t_{N_i^{\text{out}}}^{\text{exit}} = \max(d_{N_i^{\text{out}}}^{\text{exit}}, s_{N_i^{\text{out}}}^{\text{exit}}) \quad (4)$$

The implementation of demand and supply for the entry and exit of vehicles at link boundaries is quite similar to the formulation of trip-based NFD modeling approach for region-level traffic dynamics described in [58]. In this study, the formulation is applied to link-level traffic modeling instead. The update mechanism designed for these two variables resembles the derivation of receiving and sending functions of macroscopic LTM but in a mesoscopic format. While macroscopic LTM calculates the number of vehicles (flow) that can be transmitted to the next link at every discrete time step, as documented in Appendix A, LIFT determines when each individual vehicle is allowed to enter the next link based on its exit request time and the arrival time of the next backward traveling space in the downstream link at every event step.

The following summarizes the numerical procedure of the event-based resolution scheme of LIFT. These steps are performed at every event step during the simulation.

1. Initialize all vehicle entry requests at every inflow origin node by assigning the desired entry time, path ID, and the first link length as the remaining link travel distance. Set the entry and exit supply times to a negative value.
2. For all links, update the exit demand time using Eq. (2).
3. For all links, determine the next actual vehicle entry and exit times according to the logic in Eq. (4).
4. For all links, determine the next event step and step size by finding the event with the minimum (closest) vehicle entry, vehicle exit, or space arrival time. Note that it is possible to have multiple events on multiple links taking place at the same event step.
5. For all existing vehicles and backward traveling spaces, update their remaining link travel distances based on the free-flow speed V^f , backward space speed V^{space} , and the event step size determined in step 4.
6. For each link, insert and/or remove a vehicle and/or space according to the events determined in step 4.
7. For all links, update the entry and exit supply times using Eqs. (1) and (3).
8. Return to step 2 until the simulation time period ends or all the vehicles have finished their trips.

It should be noted that, unlike the event-based scheme of the mesoscopic traffic models in the commercial tools, the switch between green and red signal phases does not need to be treated as an event in step 2 since it is considered when determining the exit demand time in Eq. (2). A vehicle will only make an exit request from a link when the downstream signal phase is green.

The steps introduced above provide a rather intuitive understanding of the model. As the vehicle exit demand time and backward space arrival time actually only depend on the link length L_i , free-flow speed V^f , and backward space speed V^{space} , which are all known information, step 5 can be integrated into step 6 by calculating and storing these two time values at the link boundaries in advance whenever a vehicle or space is inserted into the link. By doing so, although the number of events to be executed would increase, we can potentially speed up the simulation since the process does not need to traverse through all the existing vehicles and spaces to update their locations. Note that the arrival of spaces is also a type of event that would be identified in step 4 in this case. This is especially beneficial for a network with a large number of circulating vehicles. LIFT is less sensitive to the number of vehicles in the network compared to microscopic simulation. Algorithm 1 explains the modeling logic in a more detailed way with a pseudocode.

Before the code begins, it is assumed that all entry requests have been initialized and stored in the *link.entry_demand* list. Every link contains a *vehicles* list and a *spaces* list to store the downstream exit demand time of vehicles and upstream arrival time of spaces. *VEH TT* and *SPACE TT* denote the link travel time required by a vehicle and a backward space, respectively. Each event type is represented by a string tag (“entry”: vehicle entry, “exit”: vehicle exit, “arrival”: space arrival).

Note that entry events only take place if a vehicle just enters the network. An entry event into the downstream link is executed together with the exit event. The *next_link* variable used in the pseudocode is determined by the downstream link of the next exiting vehicle on the link. As mentioned, the vehicle exit demand and backward space arrival times are calculated and included into *link.vehicles*, *next_link.vehicles*, and *link.spaces* directly so that the code does not need to update the position for all vehicles and spaces at every event step.

Algorithm 1: LIFT pseudocode

```

EVENT_TYPES ← ["entry", "exit", "arrival"];
time ← 0;
while time ≤ TIME do
    event_links ← [];
    for link ∈ LINKS do
        link.next_events ← [];
        link.next["entry"] ← max(t, link.entry_demand[0], link.entry_supply);
        link.next["exit"] ← max(t, link.vehicles[0], link.exit_supply);
        link.next["arrival"] ← link.spaces[0];
        link.next_event_time ← min(link.next["entry"], link.next["exit"], link.next["arrival"]);
        for event ∈ EVENT_TYPES do
            if link.next[event] = link.next_event_time then
                link.next_events + [event];
            end
        end
    end
    time ← min([link.next_event_time for link ∈ LINKS]);
    for link ∈ LINKS do
        if link.next_event_time = time then
            event_links + [link];
        end
    end
    for link ∈ event_links do
        if "entry" ∈ link.next_events then
            link.previous_entry ← time;
            link.vehicles + [time + link.VEH_TT];
            DELETE link.entry_demand[0];
        end
        if "exit" ∈ link.next_events then
            link.previous_exit ← time;
            link.spaces + [time + link.SPACE_TT];
            next_link.previous_entry ← time;
            next_link.vehicles + [time + next_link.VEH_TT];
            DELETE link.vehicles[0];
        end
        if "arrival" ∈ link.next_events then
            DELETE link.spaces[0];
        end
    end
    for link ∈ LINKS do
        link.entry_supply ← UPDATE_ENTRY_SUPPLY(link.previous_entry, link.spaces);
    end
    for link ∈ LINKS do
        link.exit_supply ← UPDATE_EXIT_SUPPLY(link.previous_exit, next_link.entry_supply);
    end
end

```

The calculations of entry supply and exit supply times are simplified by two update functions, UPDATE_ENTRY_SUPPLY and UPDATE_EXIT_SUPPLY, in the pseudocode. The update of *link.entry_supply* requires the previous entry time and the condition of backward traveling spaces, while *link.exit_supply* depends on the previous exit time and the updated *next_link.entry_supply*. For more information, the reader is referred to the repository via this URL: <https://github.com/YCNI/LIFT.git> [59].

4. Case study design

In this work, we mainly focus on verifying the accuracy of LIFT with traffic flow theoretical properties. Two case studies are designed to test its ability to capture the congestion propagation pattern. SUMO is used to generate ground-truth microscopic traffic

simulation outcomes. This section describes the parameter setups, network layouts, and OD path compositions. In addition, an empirical validation of LIFT using trajectory data on an urban road link can be found in [Appendix D](#).

4.1. Parameter settings

In the first two case studies, there is no heterogeneity considered among drivers. The desired speed in the car-following model is $V^f = 12.5$ m/s (45 km/h). The resulting saturation flow rate q^{sat} becomes 0.625 veh/s. To consider the smaller saturation flow at the beginning of a green phase due to the start-up lost time, a smaller q^{sat} of 0.5 veh/s is used for the first time step after the green phase starts in the macroscopic models, including SaF, eSaF, and LTM. In addition, for turning paths, a smaller q^{sat} of 0.4 veh/s and hence a larger headway is applied considering the slower speed for turning maneuvers because of the radius. These values are larger than the values typically used in practice since a homogeneous driving behavior is considered. It is also worth noting that the setting of a different q^{sat} for turning vehicles is only achievable in a mesoscopic model that tracks the movement of individual vehicles, while macroscopic models can only apply a saturation flow rate to a group of vehicles (flow) without differentiating their downstream turning directions.

To better reflect the traffic environment in reality, case study 3 considers the desired speed heterogeneity among drivers. The desired speeds follow a normal distribution with a standard deviation of 1.875 m/s (6.75 km/h). In case study 4, a multi-lane condition is implemented so that the overtaking maneuvers can be included. As a result, the free-flow speed and saturation flow rate in LIFT need to be changed and are determined after several preliminary simulation trials in SUMO. [Fig. 8](#) shows the free-flow traffic states of case studies 3 and 4 obtained by simulating a road link with varying inflow demand. The free-flow speeds are obtained by fitting a linear curve representing the free-flow branch of the triangular FD. As shown, for the single-lane condition in case study 3, the desired speed heterogeneity greatly affects the average speed from a macroscopic perspective due to the moving bottleneck effect, while overtaking behavior (discretionary/tactical lane-changing) in the multi-lane condition mitigates the impact so that the average speed in case study 4 becomes slightly larger. As the impact of desired speed heterogeneity and overtaking behavior on the macroscopic flow property are quantified by the fitted free-flow speeds and saturation flow rates, these parameters would then be used for the mesoscopic simulation of LIFT in case studies 3 and 4. The resulting effect of desired speed heterogeneity on the simulation outcomes will be presented in [Section 5.4](#).

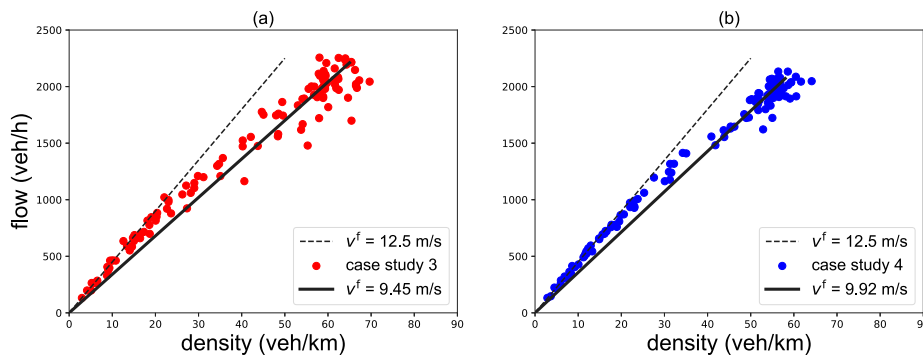


Fig. 8. SUMO simulated free-flow traffic states for (a) case study 3 and (b) case study 4 (per lane).

For all microscopic simulation cases that use SUMO in this study, its default car-following (Krauss) and lane-changing (LC2013) models are used. The time step size is set to 0.25 s instead of the typically-used 1 s to increase the accuracy of the ground-truth results since the proposed continuous-time event-based mesoscopic model can also achieve a similar level of temporal resolution. The influence of step size on the simulation outcome will be discussed at the beginning of the next section. Note that the action step length, the frequency of drivers' decision-making, remains to be 1 s. This setting not only better reflects the reality but also reduces the computation demand. On the other hand, the time step size in the compared macroscopic models is 5 s.

The start-up delay in SUMO is set to 0.75 s, which leads to a reaction time r of approximately 1 s. The effective vehicle length l^{eff} is set to 7.5 m, the summation of 5 m vehicle length and 2.5 m standstill gap. Accordingly, the speed of a backward traveling space V^{space} becomes 7.5 m/s following the calculation mentioned in the previous section. [Fig. 9](#) plots the triangular FD that reflects the determined traffic flow parameters in the first two case studies when there is no desired speed heterogeneity. [Table 2](#) summarizes the LIFT parameters in every case study.

Table 2
LIFT parameters.

Parameter	Cases 1 & 2	Case 3	Case 4
Free-flow speed V^f [m/s]	12.50	9.45	9.92
Saturation flow q^{sat} [veh/s/lane]	0.625		0.57
Effective vehicle length l^{eff} [m]		7.5	
Backward space speed V^{space} [m/s]		7.5	

It is worth noting that the speed of backward traveling spaces should theoretically be identical to the backward wave speed shown on a triangular FD, which was the setup in [\[51\]](#), but it may be different in this work. The wave speed captures the moment

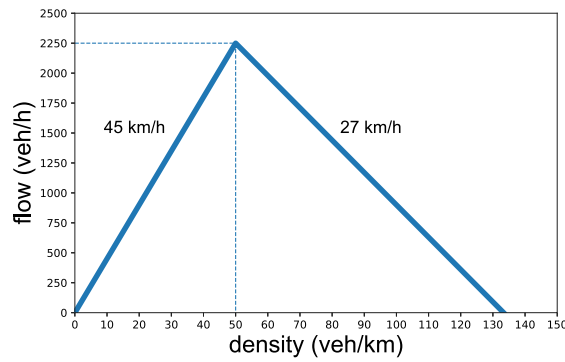


Fig. 9. Triangular FD for parameter settings in case studies 1 and 2.

when the flow of a certain location attains the queue discharge rate. However, considering a spillback condition in SUMO, the space at the upstream boundary becomes available immediately after the last queueing vehicle starts moving. Hence, using the macroscopic wave speed as the space travel speed in LIFT can result in an underestimation of traffic performance compared to theoretical ground-truth obtained from microscopic simulation. Still, the parameter may be calibrated differently depending on the real-world condition. On the other hand, this discrepancy can also be interpreted as a characteristic of interrupted traffic flow dynamics. The frequent stop-and-go movements during the accumulation and dissipation of the queue on an urban road link due to traffic signal control amplify the traffic state unsteadiness [43], which cannot be properly described with a typical triangular FD.

The parameter setups for Eissfeldt's model (the mesoscopic simulation function in SUMO) in this work can be found in [Appendix B](#).

4.2. Network layouts, OD paths, and demand profiles

A case study on a one-way single-lane corridor with a homogeneous link length of 125 m, as illustrated in [Fig. 10](#), is first designed to evaluate the model performance. The link length results in a free-flow link travel time of 10 s and a backward space link travel time of around 15 s, which are equivalent to two and three time steps in the macroscopic models, respectively. This enables the comparison of LIFT with the macroscopic models.

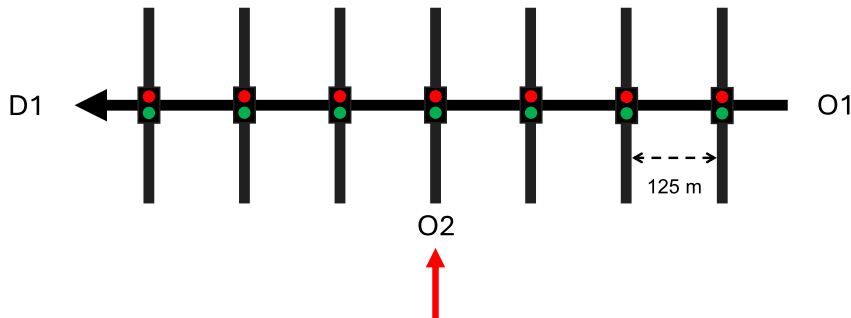


Fig. 10. Corridor and OD nodes designed in case study 1.

The signals along the corridor have a uniform green length G of 25 s and a cycle length C of 60 s. A certain degree of signal coordination is designed by setting the downstream offset to half of the free-flow link travel time $L_i/V^f/2$, which is 5 s in this case. The main traffic enters from O1 and exits at D1. Links are numbered from right to left. There is no endogenous bottleneck since the lane configuration and green-to-cycle ratio are both identical along the corridor. The disruption of traffic flow on the corridor is resulted from a turning path that joins from the minor road at the fourth intersection (upstream of link 4). The impact of turning on network traffic performance was showcased in [60]. The turning path is expected to generate long vehicle queues on link 4. The spillback would then spread upstream, affecting links 3, 2, and 1 sequentially until the low inflow demand resolves the congestion.

As all vehicles from O2 will enter the corridor and exit at D1 together with the major demand, this scenario can be modeled in a single-commodity setup without the need to differentiate vehicles using different paths.

In case study 2, we further extend the case study to the network-scale so that the influence of spillback onto the upstream vertical links with turning paths can also be tested. The purpose is also to test the ability of LIFT in tackling the FIFO condition when there are more than one exiting OD paths. A 2×4 one-way single-lane network grid is designed with a uniform link length of 125 m, as shown in [Fig. 11](#). [Table 3](#) lists all OD paths in the network grid of case study 2. The traffic signal timing plan follows the same setting mentioned in case study 1 instead for a bottleneck intersection right before nodes D1 and D6 (highlighted in red). The green

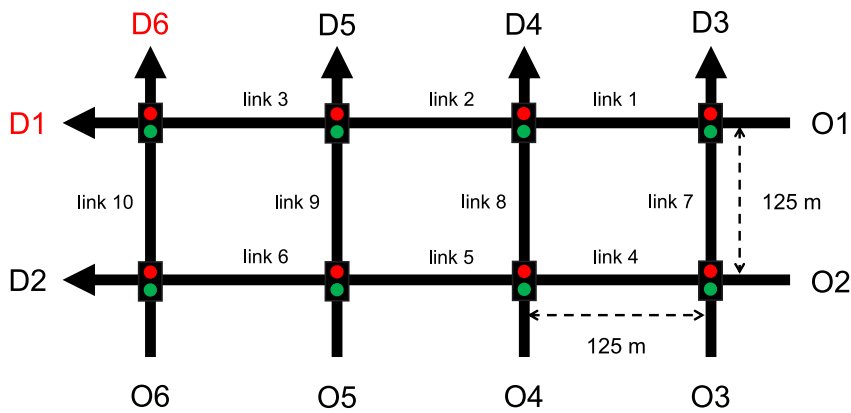


Fig. 11. Network grid and OD nodes designed in case study 2.

Table 3
OD paths in case study 2.

Path	OD	Link sequence	Path	OD	Link sequence
1	O1 - D1	1-2-3	8	O4 - D4	8
2	O2 - D2	4-5-6	9	O4 - D6	5-6-10
3	O2 - D6	4-5-6-10	10	O5 - D1	9-3
4	O3 - D1	7-1-2-3	11	O5 - D5	9
5	O3 - D3	7	12	O5 - D6	6-10
6	O3 - D6	4-5-6-10	13	O6 - D6	10
7	O4 - D1	8-2-3			

length G of the traffic signal timing plan at the downstream of links 3 and 10 is shortened from 25 s to 20 s. It is anticipated that the queue building up on these two links due to the shortened green length will spill over to their upstream links.

A three-hour scenario with varying inflow loading is implemented to mimic the onset and recovery of congestion during a typical peak period. For case study 1, the inflow loading profiles of the OD paths can be found in Fig. 12. The three-hour period is divided into twelve 15-minute time windows, while each window has a number of entry requests. The demand on path 2 (O2-D1) is 3/5 of the base demand on path 1 (O1-D1). The inflow loading profiles for case study 2 follow a similar setup but are too complex to be explained in this section. They can be found in Appendix C.

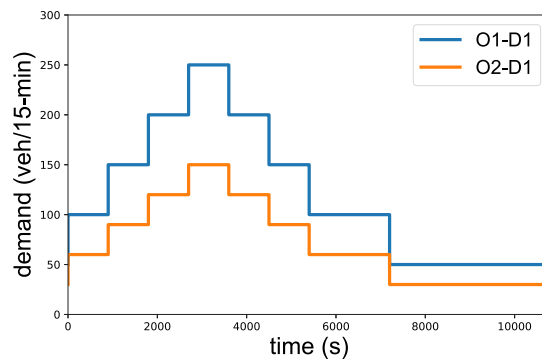


Fig. 12. Inflow loading profiles of the OD paths in case study 1.

In case study 1, a uniform entry interval (headway) is implemented based on the number of entry requests in each window. For case study 2, on the other hand, the arrival pattern of different commodities is influential. Fig. 13 uses a situation at a T-intersection to describe the importance of the entry sequence on a link. When there is a spillback condition on one of the downstream links, whether blockage to the upstream link would occur or not depends on the downstream link of the next exit vehicle, as considered by Eq. (3). This affects the delay experienced by the vehicles in the upstream link. This problem is particularly critical for urban roads with limited space. It is referred to as “diverge blocking” in [37]. Hence, to account for the stochasticity of the arrival pattern in both LIFT and SUMO, the entry intervals in case study 2 follow exponential distributions with the mean values computed based on the number of entry requests in each window. By doing so, the entry sequence of vehicles with different OD paths (commodities) is randomized rather than following a fixed pattern. Five simulation runs are conducted with different random seeds.

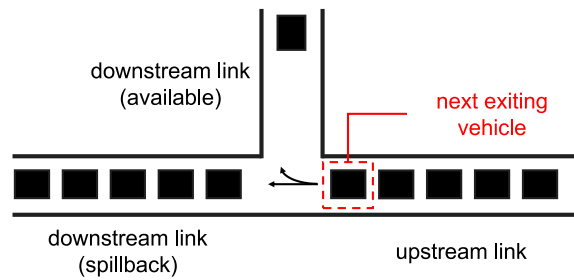


Fig. 13. Example of a diverge blocking situation.

In case studies 3 and 4, the corridor layout of case study 1 is used. The change is that the demand is down-scaled considering the reduced free-flow speed and saturation flow rate due to the desired speed heterogeneity. Also, we transform the road links from single-lane into two-lane in case study 4.

5. Results

This section presents the case study results and discusses the performance of every tested model in terms of accuracy by examining the density evolution on each link and the mean speed evolution of each path. Note that the mean path speeds are computed using the recorded travel times when vehicles enter the network without taking into account the vertical queueing time outside of the boundary so that the measurement can align with the E3 detector in SUMO. Both indicators are aggregated for every 5-minute interval.

Besides the rather qualitative comparison by looking at the evolution patterns, the mean absolute error (MAE) of link density of every link over the simulation time period is computed to quantify the deviation between the models applied and SUMO microsimulation ground-truth in case study 1, as summarized in Table 4.

5.1. Comparison with the macroscopic models

Fig. 14 shows the average density on each link in the peak period in case study 1 and can help us to understand the congestion pattern along the corridor. As can be seen from the colors, link 4 is the first to become congested due to the inflow of path 2. The queue spillback then propagates upstream to links 3, 2, and 1, increasing the density of those links. Links 5 and 6 are not influenced as they are at the downstream of the bottleneck. The congestion recovery begins when the demand of path 2 decreases so that the waiting queue on link 4 starts becoming shorter.

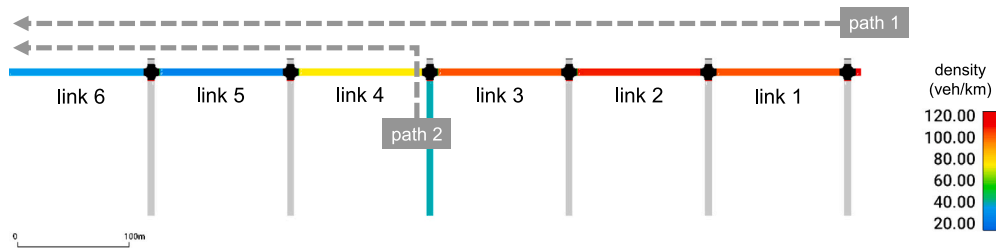


Fig. 14. Average link densities at the interval between 3600 s and 3900 s in case study 1 shown on SUMO graphical user interface.

Before showing the outcomes of the macroscopic models and LIFT, the influence of step size in the microscopic traffic simulation using SUMO is first discussed. Ideally, microsimulation should also follow the CFL condition. Otherwise, a relatively large time step size may cause a smaller outflow than the saturation flow due to the inconsistency between the minimum headway (saturation flow rate) and time step size.

Fig. 15 plots the SUMO outcomes with time step sizes from 1 s to 0.125 s. The difference between different time step sizes can easily be observed during the recovery from congestion. On all links, the settings of 1 s and 0.5 s show significantly slower queue dissipation. In these cases, the corridor cannot be recovered back to the initial undersaturated state within the simulation period. By further reducing the time step size to 0.25 s and 0.125 s, the queue can be completely dissipated, and the outcomes also converge. This implies the importance of setting a step size that is sufficiently small to avoid any numerical problem while not exceedingly small to avoid a high computation cost. Hence, the time step size in SUMO is set to 0.25 s in the following case studies for comparison with LIFT.

The results of case study 1 with a uniformly-distributed departure headway in every 15-minute interval using the macroscopic models and LIFT are shown in Fig. 16. It is found that the original SaF model slows down the onset of congestion due to the

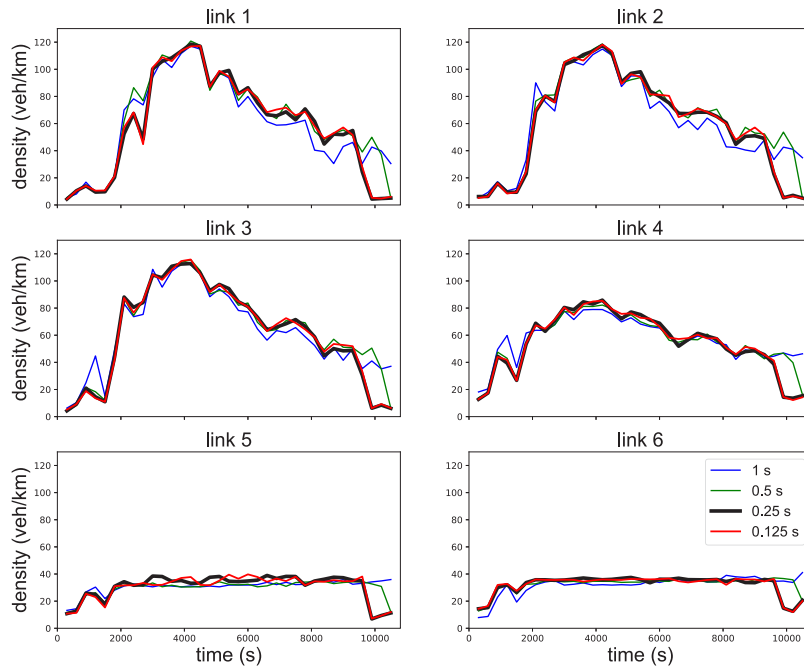


Fig. 15. Link density evolutions in case study 1 using SUMO with different time step sizes.

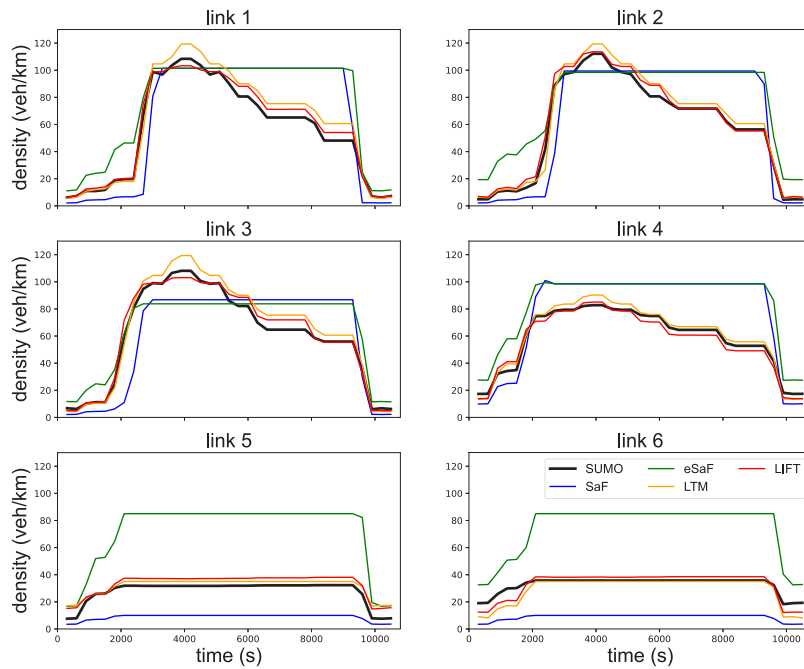


Fig. 16. Link density evolutions in case study 1 using different macroscopic models.

misconsideration of link travel time, while the eSaF model accelerates the accumulation of vehicles because of the overestimation of available downstream link space, as mentioned in Section 2.1. The problem can easily be found by looking at the density evolution at the beginning of the simulation on every link and the overall density on links 5 and 6. The density values on links 1–4, which are affected by the queue spillback, shown by both SaF and eSaF are also significantly larger than SUMO ground-truth during the congestion dissipation. This is because the inflow and outflow are balanced, implying that the queue spillback is not properly modeled.

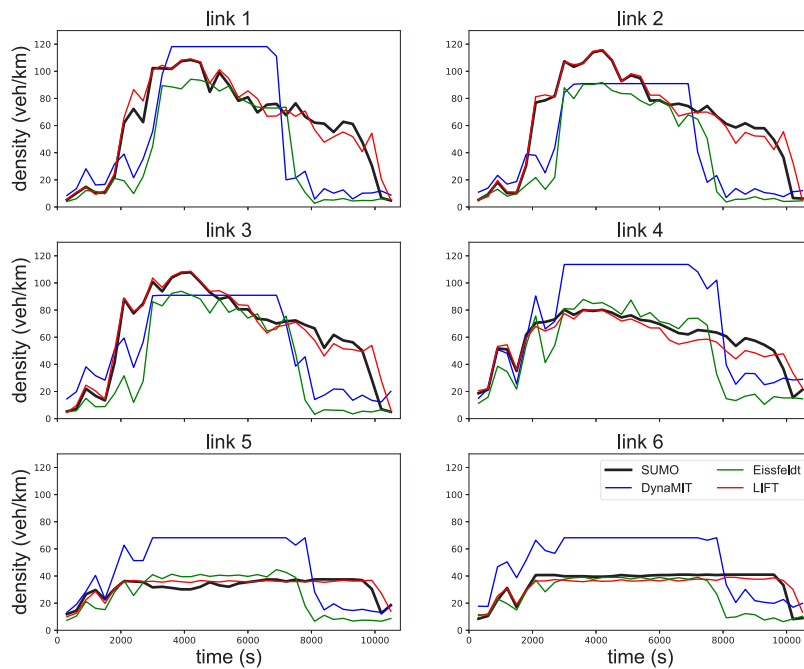


Fig. 17. Link density evolutions in case study 1 using different mesoscopic models.

On the other hand, LTM manages to capture the general pattern of density evolution thanks to its consideration of backward wave speed compared to the two SaF models. The only major difference is that there is a slight overestimation of maximum densities, which is caused by the discrete-time formulation of macroscopic LTM. This problem is comparatively minor compared to the inaccuracy shown by the SaF models. Comparatively, the results of LIFT comply with SUMO ground-truth even better because of its ability to monitor spillback more accurately and the continuous-time modeling setup facilitated by the event-based resolution scheme.

The quantitative comparison of performance between different models can be found in Table 4. Both the SaF and eSaF models have MAEs of link density which are larger than 20 veh/km on many links, indicating its inability to accurately reproduce the congestion propagation. When using LTM, the average MAEs drop below 10 veh/km. The MAEs further drop below 5 veh/km when using LIFT, which shows its exceptional performance in modeling link-level traffic dynamics along the corridor.

5.2. Comparison with the mesoscopic models

In this subsection, we further compare the performance of LIFT with the other two mesoscopic models in a scenario with exponentially-distributed departure headways. As can be seen in Fig. 17, DynaMIT exhibits a pattern similar to that shown by the eSaF. The maximum density is larger than ground-truth on links 1 and 4, where the inflow demand enters, due to the overestimated available link space. This implies the accommodation of more vehicles than the amount that can be stored in the actual available space. Consequently, earlier congestion dissipation can be observed on all links due to the underestimated influence of congestion on the vehicle progression between links. The plateaus of link density evolutions during congestion also explain the erroneous queue tail location update.

As mentioned in Section 2.2, Eissfeldt's model falsely speeds up vehicle progression because of the problematic update of exit headway to a downstream full link. This delays the occurrence of congestion propagation and accelerates congestion dissipation. Such phenomena can be observed from the results in Fig. 17. On links 1–3, congestion occurs later and is resolved earlier. The maximum density attained on these links is also smaller than ground-truth. On link 4–6 where there is no downstream spillback queue, the problem is minor at the onset of congestion, but the density still decreases to the minimum sooner. This is because vehicles in general finish their trips earlier in the simulation period so that the corridor recovers from congestion faster.

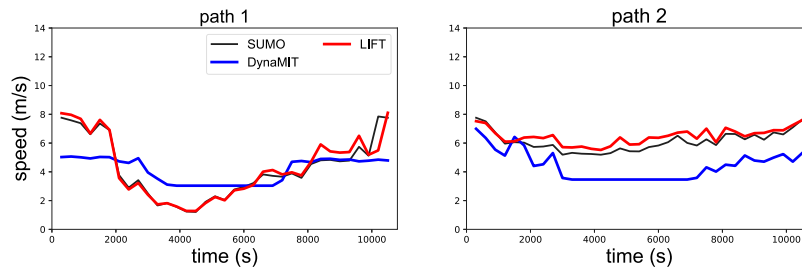
The advantage of Eissfeldt's model over DynaMIT is that it does not have that much of a problem in uncongested conditions. The quantitative assessment of the mesoscopic modeling outcomes is summarized in Table 4.

Fig. 18 plots the evolution of the mean path speed during the simulation period. The mean speed values are unable to be retrieved from Eissfeldt's model from the mesoscopic simulation function in SUMO. It can be seen that DynaMIT underestimates the congestion influence for path 1 and overestimates it for path 2. However, LIFT successfully captures the magnitude and duration of the speed drop for the two paths during congestion. The slight overestimation shown by the mean speed of path 2 is likely due to the misconsideration of detailed vehicle dynamics during acceleration. Using the desired speed in the free-flow condition slightly underestimates the actual link travel time.

Table 4

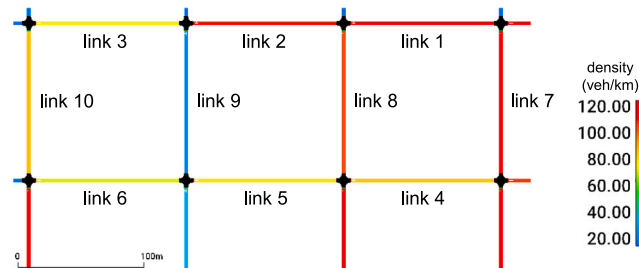
MAEs of average density (veh/km) on every link in case study 1.

Link	Macroscopic			Mesoscopic		
	SaF	eSaF	LTM	DynaMIT	Eissfeldt	LIFT
1	21.9	18.5	9.3	28.3	22.59	5.6
2	22.7	19.3	8.6	23.5	23.80	3.9
3	17.5	12.6	8.4	18.9	22.14	4.3
4	25.1	24.8	6.2	25.7	14.45	5.0
5	22.9	40.4	2.8	22.8	11.05	2.5
6	26.2	38.9	5.7	21.9	8.57	3.7

**Fig. 18.** Mean path speed evolutions in case study 1 using DynaMIT, LIFT, and SUMO.

5.3. Performance with diverge blocking phenomena

Case study 2 is designed to help demonstrate the ability of LIFT to handle FIFO in situations with diverge blockings between vehicles with different OD paths (commodities) particularly in congested conditions. The evolution pattern of link densities and path-mean speeds hence become different among scenarios with different random seeds. Therefore, for both the link densities and mean path speeds, the range between the maximum and minimum values at each aggregated time interval would be shown.

**Fig. 19.** Average link densities at the interval between 3000 s and 3300 s in case study 2 shown on SUMO graphical user interface.

We first investigate the overall congestion pattern by looking at the SUMO outcome shown in Fig. 19, while Fig. 20 plots the density evolution patterns of every link. According to the SUMO outcomes, we can see that long-lasting congestion occurs primarily on links 1–3 and 10, where paths 1 and 10 that have the highest demand pass through. Due to the congestion on link 10 directly caused by the bottleneck and those OD paths which go from link 6 to link 10, links 4–6 also have build-up queues for a period of time, albeit shorter than the congestion duration on links 1–3. Vertical links with a large number of exiting vehicles that seek to turn to links 1–3, including links 7 and 8, also experience density growth due to the blockage caused by spillback on the downstream horizontal links. The queue spillback from the bottleneck also affects the density of link 9 but is minimal due to the relatively mild congestion on link 3, where the congestion starts, as can be seen from its smaller maximum link density compared to links 1 and 2.

Looking at the density evolution patterns produced by LIFT in Fig. 20, we can see that it successfully captures the pattern of all link densities in the network grid. The maximum density on each link is quite close to the ground-truth situation.

As shown in Fig. 21, the drop and recovery of mean path speeds also align with the ground-truth pattern remarkably. In particular, the fluctuation of speeds on those paths that go through the vertical links caused by the stochastic entry sequence of different commodities can also be closely reproduced by LIFT. The slight overestimation shown on paths 5, 8, and 11 may be caused by misconsideration of acceleration dynamics, the relatively small number of samples, and the relatively short amount of time these vehicles spend in the network (short path length).

With these two case studies, we show the problems of several macroscopic and mesoscopic models proposed in the past and the ability of LIFT to accurately capture the evolution of link traffic states and path-level performance.

Besides simulation accuracy, the required computation time compared to microsimulation is also important for future studies related to urban traffic management to apply LIFT. For the scenario in case study 1, the run time required by SUMO microsimulation

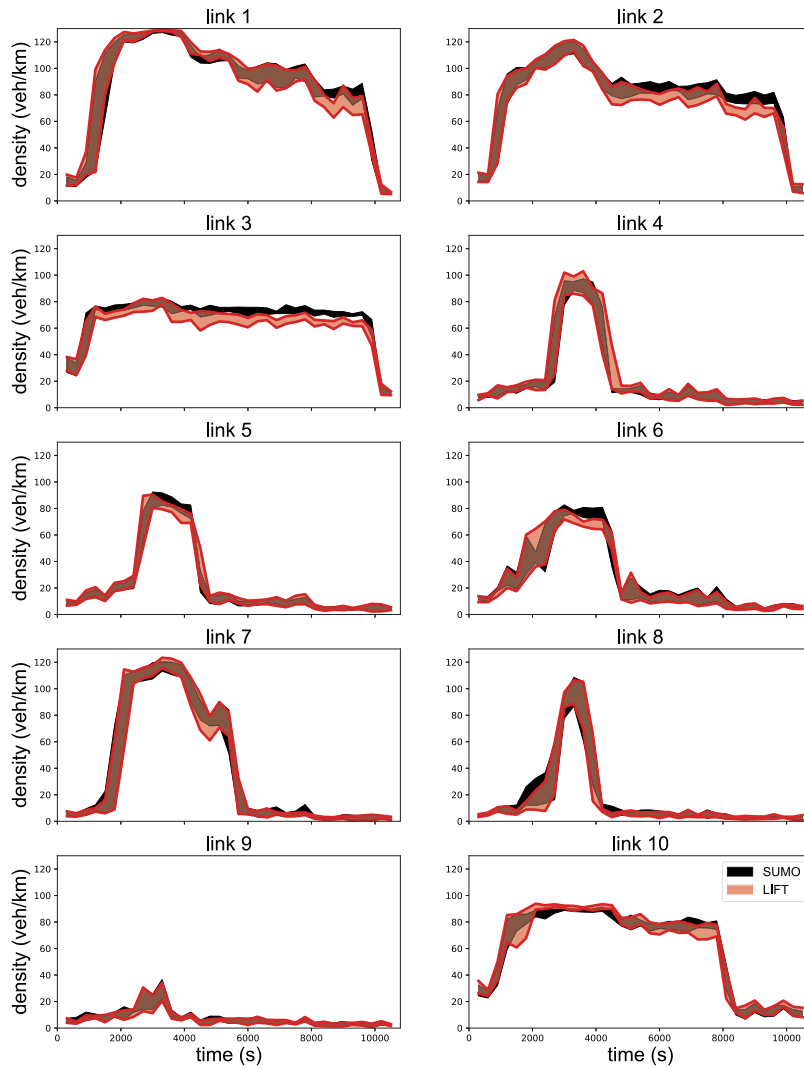


Fig. 20. Link density evolutions of case study 2 using LIFT and SUMO.

simulation is approximately 4 s with the output data storage considered on a device with a 4-core Intel Core i7 CPU running at 2.80 GHz, while it takes LIFT around 1.2 s, which is more than three times faster than SUMO. For the scenario in case study 2, which has more inflow demand and a larger network than in case study 1, SUMO needs approximately 8 s, while LIFT only spends 2.1 s. The slightly greater difference explains that the improvement on computation efficiency would become even more significant in scenarios where there are more vehicles co-existing in the network at the same moment, namely when it is more congested, or simply when the network is larger. This is because microscopic simulation has to update the position and speed for all vehicles before proceeding to the next step, LIFT only iterates through all links in the network regardless of the number of vehicles. This again demonstrates the superiority of LIFT for applications in control problems that demand real-time decision-making or optimization problems that contain numerous simulation runs. The implementation of LIFT for case study 2 can be found in [59].

5.4. Performance with desired speed heterogeneity

Figs. 22 and 23 show the impact of the desired speed heterogeneity in case studies 3 and 4 by comparing the link density evolution with that of the scenario with a homogeneous desired speed. As vehicles with a relatively slow desired speed affect the saturation flow rate and free-flow speed along the corridor, one can observe that link density increases significantly after considering heterogeneity in the single-lane condition in case study 3. It also takes longer for congestion to resolve. These imply that speed heterogeneity is indeed a crucial factor influencing congestion propagation.

On the other hand, the negative impact is greatly mitigated in the multi-lane condition in case study 4. Maximum link densities are closer to the scenario with a homogeneous desired speed although there is still a delayed congestion recovery. This is because

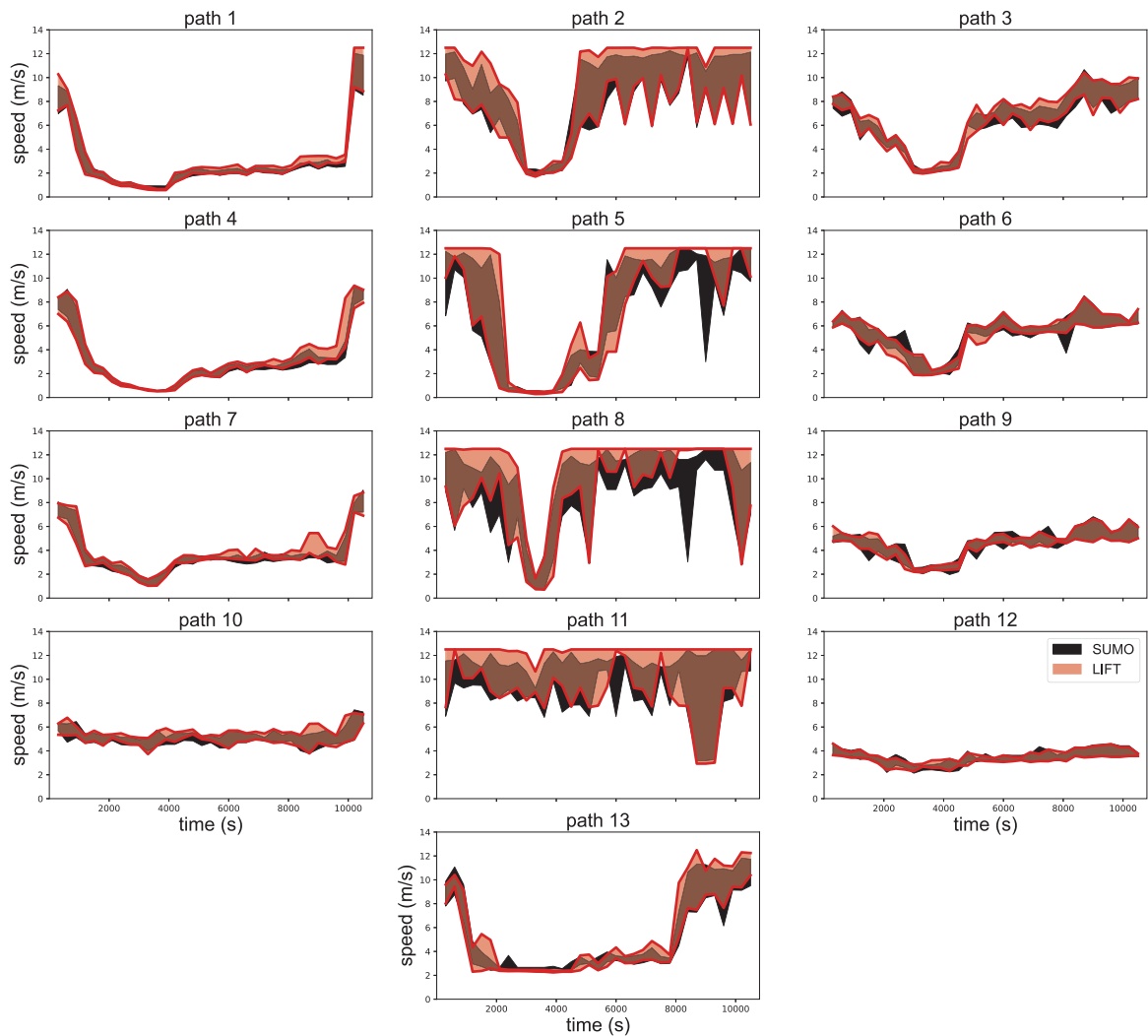


Fig. 21. Mean path speed evolutions of case study 2 using LIFT and SUMO.

fast vehicles can utilize the additional lane to overtake those slow vehicles and reduce the moving bottleneck effect. Therefore, the free-flow speed would not decrease drastically.

To demonstrate the ability of LIFT to replicate traffic dynamics in scenarios with desired speed heterogeneity, the scenarios of case studies 3 and 4 are simulated with updated LIFT macroscopic parameters. Link density evolutions are shown in Figs. 24 and 25.

As can be seen, LIFT is able to reproduce the traffic dynamics along the corridor even when the desired speed heterogeneity is considered simply by using the well-calibrated macroscopic parameters. The maximum link densities and the onset and recovery of congestion queues align fairly well with ground-truth microscopic simulation outcomes in both case studies.

6. Concluding remarks

This study first reviews link-level traffic flow models used in the past and points out their problems that are particularly influential for urban road networks considering the dynamics exhibited by interrupted flow due to traffic signal control, i.e., the frequent accumulation and dissipation of waiting queues on a road. Second, LIFT, an implementation of the macroscopic LTM concept at the mesoscopic level specifically for urban traffic flow, is developed. The key advantage of LIFT in comparison to other models is its ability to monitor queue spillback on an urban link by utilizing the concept of backward traveling spaces. With the event-based resolution scheme, we enable continuous-time modeling for multi-commodity flow with fully-ensured FIFO in diverge blocking situations.

The theoretical validity of the model, which was rarely discussed before in previous studies that have similar purposes, is examined by comparing with the outcomes generated using SUMO microscopic traffic simulation. We also evaluate the performance

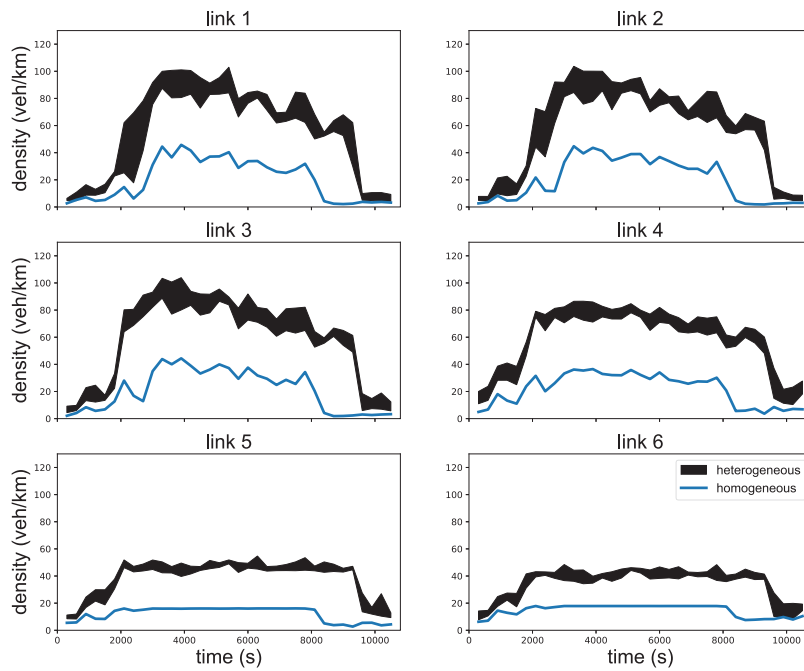


Fig. 22. Impact of desired speed heterogeneity in case study 3 (single-lane).

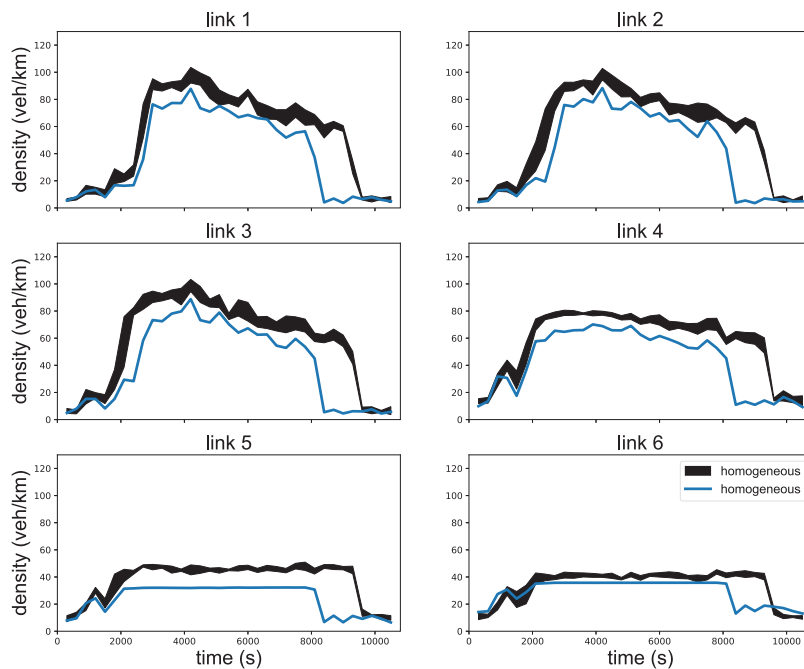


Fig. 23. Impact of desired speed heterogeneity in case study 4 (multi-lane, per lane density).

of LIFT and compare it with the other models in one of the case studies. According to the link density evolutions analyzed, the case studies show that LIFT can accurately reproduce the congestion propagation pattern along a signalized corridor and within a network grid. It outperforms the SaF models and DynaMIT by solving the problem of overestimated available downstream link space. The proper consideration of backward traveling spaces also provides a way to update the entry and exit conditions at link boundaries more appropriately compared to Eissfeldt's model, which is currently being used as the mesoscopic modeling function in SUMO. In addition, a validation with empirical trajectory data on a lane in a real-world urban road is performed and can be found in [Appendix](#)

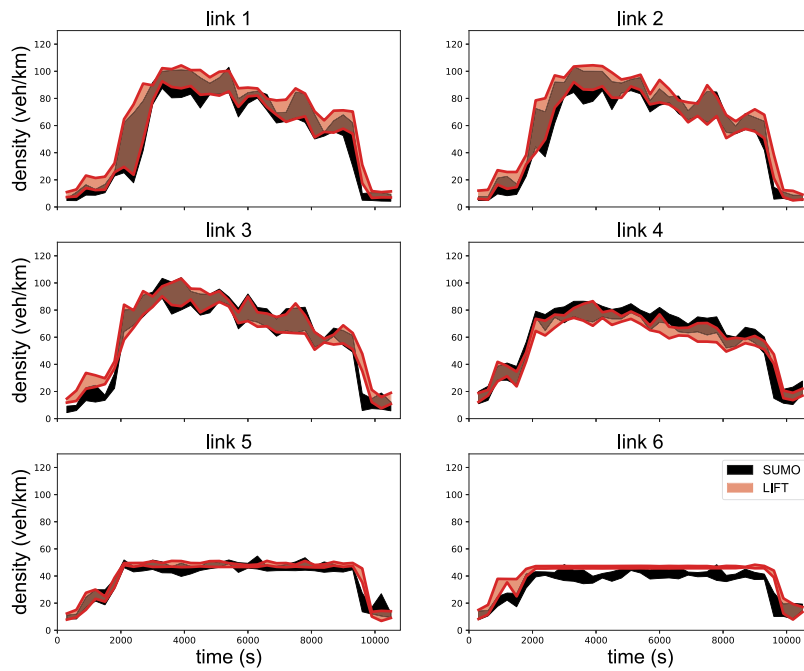


Fig. 24. Link density evolutions of case study 3 using LIFT and SUMO.

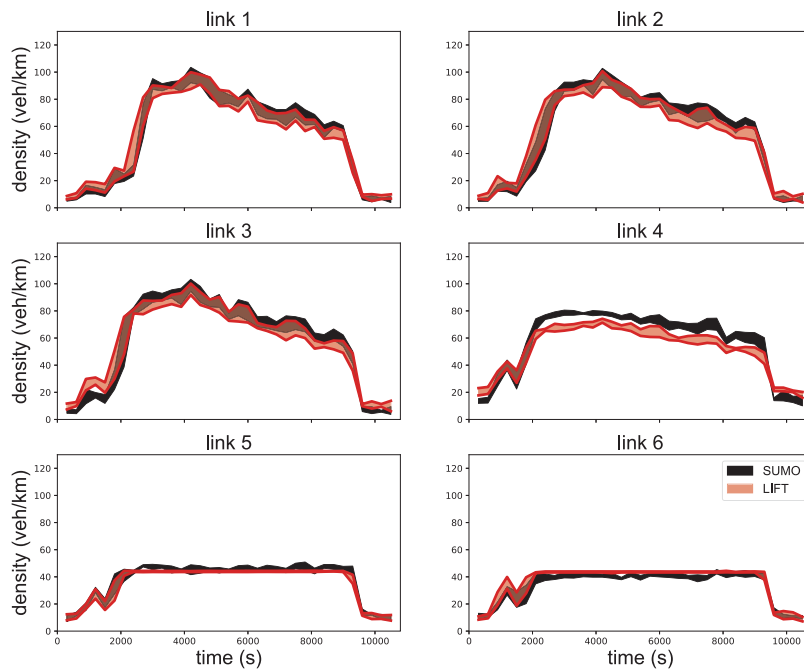


Fig. 25. Link density evolutions of case study 4 using LIFT and SUMO (per lane density).

D. The occurrences of queue spillback in every signal cycle can be correctly replicated with LIFT. Based on these results, we can conclude that it is important to design a mesoscopic traffic flow model that carefully follows the principle of the LWR model or its discrete-time correspondence, such as CTM and LTM.

The case study results also demonstrate that it is possible for a mesoscopic-level modeling approach to obtain outcomes that are sufficiently close to the microscopic simulation output for an urban road network given a fixed signal timing plan even in

scenarios with driving behavioral heterogeneity considered. In addition, the ability to capture the mean speed of OD paths at every time interval allows the model to incorporate path choice behavior for each individual vehicle/traveler considering the travel time affected by congestion in a DTA framework.

LIFT can be regarded as the discrete-event interrupted flow version of the discrete-time model proposed in [52]. While they aimed at replicating the macroscopic continuous counterpart, LIFT is validated against microscopic simulation output. Therefore, many types of microscopic behavioral heterogeneity, such as the relatively long headway between turning movements, can be better implemented into such a model formulation so that the modeling accuracy can be significantly improved.

In terms of computation efficiency, LIFT can be considered an alternative that lies between the mesoscopic simulation function in commercial traffic simulation tools that still considers the interaction between individual vehicles with simplified car-following models and macroscopic link-level traffic modeling approaches that model the propagation of aggregated flow. For the traffic performance assessment of large-scale urban road networks that contain multiple ODs and are heavily congested due to frequent queue spillback phenomena, the computation requirement can be significantly reduced when using LIFT compared to choosing microscopic simulation, while the accuracy can still be preserved to a great extent. The advantage on computation efficiency makes LIFT a suitable option for optimization problems that take traffic into account, e.g., network design, multi-modal road space allocation, and real-time routing strategy, etc.

In this study, mandatory (strategic) lane-changing within a road link for the downstream turning direction is not considered in the design of LIFT. We assume that vehicles would conduct lane-selection before entering the link, as modeled in the mesoscopic function of Aimsun [55]. However, this excludes the potential disruption caused by lane-changing in the middle of a link. To enhance the reliability of LIFT in multi-lane urban road networks, future work can focus on capturing the influence of mandatory lane-changing behaviors on the efficiency of interrupted traffic flow on a road link [61]. Furthermore, the extension by incorporating more sources of heterogeneity and uncertainty in a multi-modal urban road network with buses and bicycles can also be considered. This is beneficial for the applicability of LIFT in various strategic-level or policy-making problems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Macroscopic LTM

The formulation of the original macroscopic LTM for single-commodity traffic flow modeling with traffic signals proposed in [20] is elaborated in this section. Eqs. (A.1) and (A.2) are the sending and receiving functions which show the maximum number of vehicles that can exit and enter the link. N_a^0 and N_a^L represent the cumulative number of vehicles passing through the upstream and downstream ends of the link a . τ_a^f and τ_a^w denote the time steps required for a vehicle and a backward traveling space to traverse through the link, respectively. These two equations resemble the update of demand and supply for entry and exit in LIFT, as written in those equations in Section 3 of the main text. The main difference is that LIFT determines the movement between links for individual vehicles instead of aggregated flow.

$$S_a(t) = \min(N_a^0(t+1 - \tau_a^f) - N_a^L(t), q^{\text{sat}}) \quad (\text{A.1})$$

$$R_a(t) = \min(N_a^L(t+1 - \tau_a^w) - N_a^0(t), q^{\text{sat}}) \quad (\text{A.2})$$

The outflow from link a to link i is controlled by a binary indicator η_{ai} , showing whether the traffic signal from link a to link i is green, and determined by the minimum between the number of vehicles which desire to go from link a to link i and the available space (receiving function) on link i , as expressed in Eq. (A.3). θ_{ai} denotes the split ratio of vehicles going to link i among all vehicles on link a . Eqs. (A.4) and (A.5) then calculate the total number of vehicles entering the link and exiting the link.

$$u_{ai}(t) = \eta_{ai} \cdot \min(S_a(t) \cdot \theta_{ai}, R_i(t)) \quad (\text{A.3})$$

$$q_a^{\text{in}}(t) = \sum_{i \in I(a)} u_{ia}(t) \quad (\text{A.4})$$

$$q_a^{\text{out}}(t) = \sum_{i \in O(a)} u_{ai}(t) \quad (\text{A.5})$$

Eqs. (A.6) and (A.7) update the cumulative numbers of vehicles which pass through the upstream and downstream ends of the link, respectively, which are used in the computation of sending S and receiving R functions.

$$N_a^0(t+1) = N_a^0(t) + q_a^{\text{in}}(t) \quad (\text{A.6})$$

$$N_a^L(t+1) = N_a^L(t) + q_a^{\text{out}}(t) \quad (\text{A.7})$$

Appendix B. Parameters for the mesoscopic model in SUMO

Table B.1 shows the parameters specified for the mesoscopic simulation function in SUMO which is based on Eissfeldt's model [46]. Other parameters that are not mentioned here follow the default setting in SUMO. Note that edgelenh needs to be carefully selected otherwise an inappropriate spatial discretization may largely underestimate the jam density. The edge state is determined based on the critical density shown on an FD if jam-threshold is set to -1 . Instead of using any continuous node modeling approach to approximate traffic signal control by changing TLS-Flow-Penalty, the explicit on-and-off logic is applied.

Table B.1
Parameters used in Eissfeldt's model.

Parameter	Description	Value
edgelenh	Length of a homogeneous edge segment	50 m
tauff	Factor for calculating the net time headway between two free edges	1.20
taufj	Factor for calculating the net time headway from a free edge to a jammed edge	1.20
taujf	Factor for calculating the net time headway from a jammed edge to a free-flow edge	1.20
taujj	Factor for calculating the time headway between two jammed edges (Time for a backward traveling space to propagate an effective vehicle length)	1.00
jam-threshold	Occupancy threshold to consider a segment jammed	-1
junction-control	Enable traffic signal timing modeling	True
lane-queue	Separate queues on different lanes	True

Appendix C. Inflow loading profiles in case study 2

Table C.1 contains the inflow loading in each time window (per 15 min) of every OD path in case study 2. Paths 1 and 13 which intersect with each other at the bottleneck node are chosen to be the paths that have the largest inflow demand so that congestion can be easily generated.

Table C.1
Inflow loading profiles of every OD path in case study 2.

Path	Base inflow loading profile (veh/15-min)
Path 1 (O1 - D1)	[50, 75, 100, 100, 100, 75, 50, 25, 25, 25, 25, 25]
Path 2 (O2 - D2)	1/5 of the base profile
Path 3 (O2 - D6)	4/5 of the base profile
Path 4 (O3 - D1)	3/5 of the base profile
Path 5 (O3 - D3)	1/5 of the base profile
Path 6 (O3 - D6)	1/5 of the base profile
Path 7 (O4 - D1)	3/5 of the base profile
Path 8 (O4 - D4)	1/5 of the base profile
Path 9 (O4 - D6)	1/5 of the base profile
Path 10 (O5 - D1)	3/5 of the base profile
Path 11 (O5 - D5)	1/5 of the base profile
Path 12 (O5 - D6)	1/5 of the base profile
Path 13 (O6 - D6)	The same as the base profile

Appendix D. Empirical validation

To validate the interrupted traffic flow dynamics exhibited by LIFT for a real-world road traffic scenario, trajectory data that contain exact knowledge of inflow demand and link outflow are required. Therefore, we decide to use the data collected by drones in the pNEUMA dataset [62] for this case study. Trajectories on the center lane of a 150-m long road link on the north-west section of Alexandras Avenue in Athens, Greece from 10:00:30 to 10:13:30 in the morning are selected, as documented in [63].

The validation of the model against trajectory data aims at showing the ability of LIFT to capture the empirical traffic dynamics, particularly the queue spillback phenomenon. Compared to other studies that calibrated and validated the OD demand, route choice behavior, and network supply parameters in a mesoscopic traffic model directly using detector count data, such as Ben-Akiva et al. [42], the objective in this work is at a finer level and should be considered essentially different. On the other hand, compared to Celikoglu and Dell'Orco [39] that only checked the outflow on a bridge and de Souza et al. [52] that used loop detector data on a motorway to validate the flow and density evolution, this approach that utilizes trajectory data allows us to observe and validate the occurrence of queue spillback by knowing the exact link traffic state.

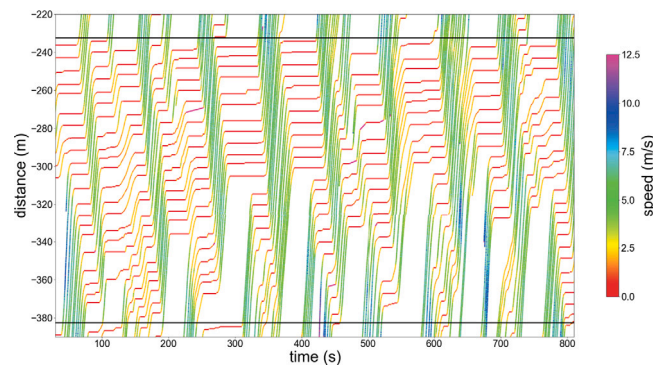


Fig. D.1. Time-space diagram of the empirical scenario.

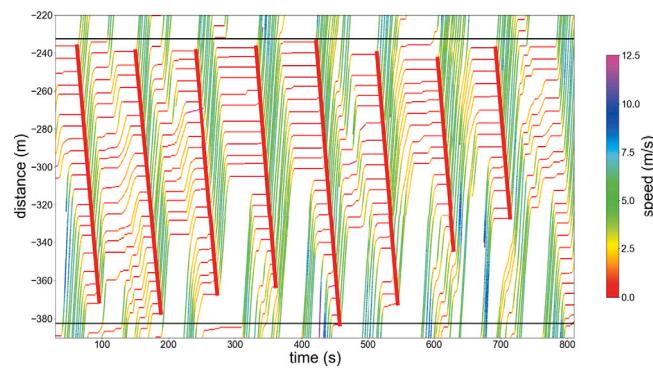


Fig. D.2. Dissipating shockwaves (red) shown on the time-space diagram of the empirical scenario.

Table D.2

LIFT parameters.

Parameter	Empirical scenario
Free-flow speed V^f [m/s]	11.5
Saturation flow q^{sat} [veh/s/lane]	0.35
Effective vehicle length l^{eff} [m]	6.5
Backward space speed V^{space} [m/s]	4.5

Fig. D.1 shows the vehicle trajectories on the lane. Vehicles that performed lane-changing to leave the lane are removed. Motorcycles are also excluded as they are mostly shown to ride near the edges of the lane, minimizing their impact on car traffic dynamics. The long queues in every signal cycle indicate that there are sufficient queue accumulation and dissipation phenomena in the scenario, making it a proper case for validation of congestion dynamics.

First, the macroscopic traffic flow parameters used in LIFT need to be determined using the trajectories. To know the maximum number of vehicles the lane can accommodate, the effective vehicle length is obtained by analyzing the queue length and number of queueing vehicles. The free-flow speed is the average maximum speed of every included vehicle. The saturation flow is calculated by taking the average of outflow in every green duration. The backward space speed is estimated by averaging the dissipating wave speeds, as shown in Fig. D.2. The vehicle entry times and signal timing plan in each cycle are prepared according to the time-space diagram. Table D.2 summarizes the parameters. One may see the notable difference of saturation flow and backward space speed with the values used in the simulation experiment in Section 4.

The validation outcome is examined by looking at the link traffic state evolution and checking if the occurrence of queue spillback phenomena can be replicated. Fig. D.3 demonstrates that the evolution of number of vehicles simulated with LIFT matches the empirical pattern. The relatively large difference at around 700 s is caused by a greater saturation outflow than the average in that particular green duration, showing the great uncertainty of urban traffic flow in reality. Furthermore, the blue curve in the figure shows the evolution of available space at the upstream end of the lane. Zero available space indicates the occurrence of queue spillback. One can see that they are successfully replicated at the moment that the accumulating shockwaves propagate to the upstream end of the lane. The available space also does not always grow as the number of vehicles on the lane decreases. Instead, a time lag exists between these two types of event because the available space should only grow whenever a backward traveling space reaches the upstream end of the link. This again verifies the ability of LIFT to capture the queue dynamics within a road link without additionally relying on optimization methods to calibrate the parameters.

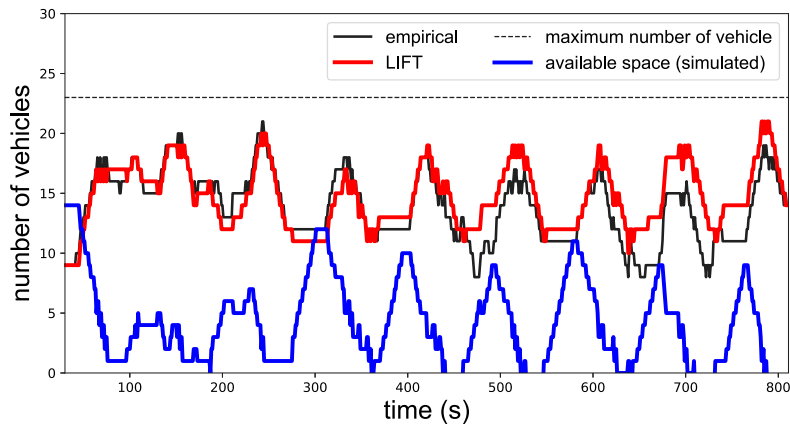


Fig. D.3. Evolution of number of vehicles and available space on the lane.

It is worth mentioning that, compared to motorway traffic, empirical validation for urban traffic flow models still faces significant challenges due to the lack of precise signal timing information, complex vehicle movements, great behavioral stochasticity, and mixed (multi-modal) road traffic environment. This case study serves as an example for such an attempt and can still be improved.

Data availability

The source code of the Python implementation of LIFT for the 2 x 4 grid network described in case study 2 in this work is openly available and can be accessed via the following link.

[LIFT: Link-level Interrupted Flow Traffic dynamics simulation \(Original data\) \(Github\)](#)

References

- [1] M. Keyvan-Ekbatani, A. Kouvelas, I. Papamichail, M. Papageorgiou, Exploiting the fundamental diagram of urban networks for feedback-based gating, *Transp. Res. B* 46 (10) (2012) 1393–1403, <http://dx.doi.org/10.1016/j.trb.2012.06.008>.
- [2] Z. Gu, M. Saberi, Simulation-based optimization of toll pricing in large-scale urban networks using the network fundamental diagram: A cross-comparison of methods, *Transp. Res. C* 122 (2021) 102894, <http://dx.doi.org/10.1016/j.trc.2020.102894>.
- [3] M. Johari, M. Keyvan-Ekbatani, L. Leclercq, D. Ngoduy, H.S. Mahmassani, Macroscopic network-level traffic models: Bridging fifty years of development toward the next era, *Transp. Res. C* 131 (2021) 103334, <http://dx.doi.org/10.1016/j.trc.2021.103334>.
- [4] V.L. Knoop, S.P. Hoogendoorn, J.W.C. Van Lint, Routing strategies based on macroscopic fundamental diagram, *Transp. Res. Rec.: J. Transp. Res. Board* 2315 (1) (2012) 1–10, <http://dx.doi.org/10.3141/2315-01>.
- [5] L. Leclercq, C. Parzani, V.L. Knoop, J. Amourette, S.P. Hoogendoorn, Macroscopic traffic dynamics with heterogeneous route patterns, *Transp. Res. C* 59 (2015) 292–307, <http://dx.doi.org/10.1016/j.trc.2015.05.006>.
- [6] G. Mariotte, L. Leclercq, Heterogeneous perimeter flow distributions and MFD-based traffic simulation, *Transp. B* 7 (1) (2019) 1378–1401, <http://dx.doi.org/10.1080/21680566.2019.1627954>.
- [7] S. Jiang, M. Keyvan-Ekbatani, D. Ngoduy, Partitioning of urban networks with polycentric congestion pattern for traffic management policies: Identifying protected networks, *Comput.-Aided Civ. Infrastruct. Eng.* 38 (4) (2023) 508–527, <http://dx.doi.org/10.1111/mice.12895>.
- [8] X. Wu, H.X. Liu, N. Geroliminis, An empirical analysis on the arterial fundamental diagram, *Transp. Res. B* 45 (1) (2011) 255–266, <http://dx.doi.org/10.1016/j.trb.2010.06.003>.
- [9] M. Saberi, H. Hamedmoghadam, M. Ashfaq, S.A. Hosseini, Z. Gu, S. Shafiei, D.J. Nair, V. Dixit, L. Gardner, S.T. Waller, M.C. González, A simple contagion process describes spreading of traffic jams in urban networks, *Nat. Commun.* 11 (1) (2020) 1616, <http://dx.doi.org/10.1038/s41467-020-15353-2>.
- [10] L. Ambühl, M. Menendez, M.C. González, Understanding congestion propagation by combining percolation theory with the macroscopic fundamental diagram, *Commun. Phys.* 6 (1) (2023) 26, <http://dx.doi.org/10.1038/s42005-023-01144-w>.
- [11] M.D. Simoni, C.G. Claudel, A semi-analytic approach to model signal plans in urban corridors and its application in metaheuristic optimization, *Transp. B* 7 (1) (2019) 185–201, <http://dx.doi.org/10.1080/21680566.2017.1370397>.
- [12] P.-E. Mazaré, A.H. Dehwan, C.G. Claudel, A.M. Bayen, Analytical and grid-free solutions to the Lighthill–Whitham–Richards traffic flow model, *Transp. Res. B* 45 (10) (2011) 1727–1748, <http://dx.doi.org/10.1016/j.trb.2011.07.004>.
- [13] C.F. Daganzo, The cell transmission model: A dynamic representation of highway traffic consistent with the hydrodynamic theory, *Transp. Res. B* 28 (4) (1994) 269–287, [http://dx.doi.org/10.1016/0191-2615\(94\)90002-7](http://dx.doi.org/10.1016/0191-2615(94)90002-7).
- [14] S.K. Godunov, A finite difference method for the numerical computation of discontinuous solutions of the equations of fluid dynamics, *Mat. Sb.* 47 (1959) 271–306.
- [15] A. Srivastava, W.-L. Jin, J.-P. Lebacque, A modified cell transmission model with realistic queue discharge features at signalized intersections, *Transp. Res. B* 81 (2015) 302–315, <http://dx.doi.org/10.1016/j.trb.2015.05.013>.
- [16] L. Adacher, M. Tiriolo, A macroscopic model with the advantages of microscopic model: A review of cell transmission model's extensions for urban traffic networks, *Simul. Model. Pr. Theory* 86 (2018) 102–119, <http://dx.doi.org/10.1016/j.simpat.2018.05.003>.
- [17] Z. Hao, R. Boel, Z. Li, Model based urban traffic control, part I: Local model and local model predictive controllers, *Transp. Res. C* 97 (2018) 61–81, <http://dx.doi.org/10.1016/j.trc.2018.09.026>.

- [18] G. Newell, A simplified theory of kinematic waves in highway traffic, part I: General theory, *Transp. Res. B* 27 (4) (1993) 281–287, [http://dx.doi.org/10.1016/0191-2615\(93\)90038-C](http://dx.doi.org/10.1016/0191-2615(93)90038-C).
- [19] I. Yperman, *The Link Transmission Model for Dynamic Network Loading* (Ph.D. thesis), Katholieke Universiteit Leuven, 2007.
- [20] I. Yperman, C. Tampère, B. Immers, A kinematic wave dynamic network loading model including intersection delays, in: *The 86th Annual Meeting of the Transportation Research Board*, Washington DC, USA, 2007.
- [21] K. Han, V.V. Gayah, B. Piccoli, T.L. Friesz, T. Yao, On the continuum approximation of the on-and-off signal control on dynamic traffic networks, *Transp. Res. B* 61 (2014) 73–97, <http://dx.doi.org/10.1016/j.trb.2014.01.001>.
- [22] K. Han, V.V. Gayah, Continuum signalized junction model for dynamic traffic networks: Offset, spillback, and multiple signal phases, *Transp. Res. B* 77 (2015) 213–239, <http://dx.doi.org/10.1016/j.trb.2015.03.005>.
- [23] R. Yahyamoazarani, C.M. Tampère, The continuous signalized (COS) node model for dynamic traffic assignment, *Transp. Res. B* 168 (2023) 56–80, <http://dx.doi.org/10.1016/j.trb.2022.12.003>.
- [24] G.S. van de Weg, H.L. Vu, A. Hegyi, S.P. Hoogendoorn, A hierarchical control framework for coordination of intersection signal timings in all traffic regimes, *IEEE Trans. Intell. Transp. Syst.* 20 (5) (2019) 1815–1827, <http://dx.doi.org/10.1109/TITS.2018.2837162>.
- [25] G.S. van de Weg, M. Keyvan-Ekbatani, A. Hegyi, S.P. Hoogendoorn, Linear MPC-based urban traffic control using the link transmission model, *IEEE Trans. Intell. Transp. Syst.* 21 (10) (2020) 4133–4148, <http://dx.doi.org/10.1109/TITS.2019.2938795>.
- [26] M.P. Raadsen, M.C. Bliemer, M.G. Bell, An efficient and exact event-based algorithm for solving simplified first order dynamic network loading problems in continuous time, *Transp. Res. B* 92 (2016) 191–210, <http://dx.doi.org/10.1016/j.trb.2015.08.004>.
- [27] W. Himpe, R. Corthout, M.C. Tampère, An efficient iterative link transmission model, *Transp. Res. B* 92 (2016) 170–190, <http://dx.doi.org/10.1016/j.trb.2015.12.013>.
- [28] D. Gazis, R. Potts, *The oversaturated intersection*, in: *Proceedings of the Second International Symposium on Traffic Theory*, London, UK, 1963, pp. 221–237.
- [29] K. Aboudolas, M. Papageorgiou, E. Kosmatopoulos, Store-and-forward based methods for the signal control problem in large-scale congested urban road networks, *Transp. Res. C* 17 (2) (2009) 163–174, <http://dx.doi.org/10.1016/j.trc.2008.10.002>.
- [30] A. Kouvelas, J. Lioris, S.A. Fayazi, P. Varaiya, Maximum pressure controller for stabilizing queues in signalized arterial networks, *Transp. Res. Rec.: J. Transp. Res. Board* 2421 (1) (2014) 133–141, <http://dx.doi.org/10.3141/2421-15>.
- [31] F. de Souza, R.C. Carlson, E.R. Muller, K. Ampountolas, Multi-commodity traffic signal control and routing with connected vehicles, *IEEE Trans. Intell. Transp. Syst.* 23 (5) (2022) 4111–4121, <http://dx.doi.org/10.1109/TITS.2020.3041436>.
- [32] S. Lin, B. De Schutter, Y. Xi, H. Hellendoorn, Efficient network-wide model-based predictive control for urban traffic networks, *Transp. Res. C* 24 (2012) 122–140, <http://dx.doi.org/10.1016/j.trc.2012.02.003>.
- [33] D. Tsitsokas, A. Kouvelas, N. Geroliminis, Modeling and optimization of dedicated bus lanes space allocation in large networks with dynamic congestion, *Transp. Res. C* 127 (2021) 103082, <http://dx.doi.org/10.1016/j.trc.2021.103082>.
- [34] L. Wei, S.T. Waller, Y. Mei, P. Chen, Y. Wang, M. Wang, A link-based flow model with turn-level queue transmission and time-varying free-flow speed for urban road networks, *Transp. Res. C* 170 (2025) 104937, <http://dx.doi.org/10.1016/j.trc.2024.104937>.
- [35] M.P. Raadsen, M.C. Bliemer, Continuous-time general link transmission model with simplified fanning, part II: Event-based algorithm for networks, *Transp. Res. B* 126 (2019) 471–501, <http://dx.doi.org/10.1016/j.trb.2018.01.003>.
- [36] H. Bar-Gera, M. Carey, Representation requirements for perfect first-in-first-out verification in continuous flow dynamic models, *Transp. Res. B* 100 (2017) 284–301, <http://dx.doi.org/10.1016/j.trb.2017.02.002>.
- [37] H. Bar-Gera, M. Carey, Constructing a cell transmission model solution adhering fully to first-in-first-out conditions, *Transp. Res. B* 161 (2022) 247–267, <http://dx.doi.org/10.1016/j.trb.2022.04.002>.
- [38] M. Di Gangi, Modelling dynamic network loading on transportation networks through a continuous packet approach, *Math. Model. Syst.* 2 (3) (1996) 175–196, <http://dx.doi.org/10.1080/13873959608837037>.
- [39] H.B. Celikoglu, M. Dell'Orco, Mesoscopic simulation of a dynamic link loading process, *Transp. Res. C* 15 (5) (2007) 329–344, <http://dx.doi.org/10.1016/j.trc.2007.05.003>.
- [40] M. Di Gangi, Modeling evacuation of a transport system: Application of a multimodal mesoscopic dynamic traffic assignment model, *IEEE Trans. Intell. Transp. Syst.* 12 (4) (2011) 1157–1166, <http://dx.doi.org/10.1109/TITS.2011.2143408>.
- [41] R. Jayakrishnan, H.S. Mahmassani, T.-Y. Hu, An evaluation tool for advanced traffic information and management systems in urban networks, *Transp. Res. C* 2 (3) (1994) 129–147, [http://dx.doi.org/10.1016/0968-090X\(94\)90005-1](http://dx.doi.org/10.1016/0968-090X(94)90005-1).
- [42] M.E. Ben-Akiva, S. Gao, Z. Wei, Y. Wen, A dynamic traffic assignment model for highly congested urban networks, *Transp. Res. C* 24 (2012) 62–82, <http://dx.doi.org/10.1016/j.trc.2012.02.006>.
- [43] J.A. Laval, Hysteresis in traffic flow revisited: An improved measurement method, *Transp. Res. B* 45 (2) (2011) 385–391, <http://dx.doi.org/10.1016/j.trb.2010.07.006>.
- [44] M. Ben-Akiva, M. Bierlaire, D. Burton, H.N. Koutsopoulos, R. Mishalani, Network state estimation and prediction for real-time traffic management, *Netw. Spat. Econ.* 1 (3–4) (2001) 293–318, <http://dx.doi.org/10.1023/A:1012883811652>.
- [45] P.A. Lopez, E. Wiessner, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötterod, R. Hilbrich, L. Lucken, J. Rummel, P. Wagner, Microscopic traffic simulation using SUMO, in: *2018 21st International Conference on Intelligent Transportation Systems, ITSC*, IEEE, 2018, pp. 2575–2582, <http://dx.doi.org/10.1109/ITSC.2018.8569938>.
- [46] N.G. Eissfeldt, *Vehicle-Based Modelling of Traffic. Theory and Application to Environmental Impact Modelling* (Ph.D. thesis), Universität zu Köln, 2004.
- [47] D. Charypar, K.W. Axhausen, K. Nagel, Event-driven queue-based traffic flow microsimulation, *Transp. Res. Rec.: J. Transp. Res. Board* 2003 (1) (2007) 35–40, <http://dx.doi.org/10.3141/2003-05>.
- [48] C. Osorio, G. Flötteröd, M. Bierlaire, Dynamic network loading: A stochastic differentiable model that derives link state distributions, *Transp. Res. B* 45 (9) (2011) 1410–1423, <http://dx.doi.org/10.1016/j.trb.2011.05.014>.
- [49] C. Osorio, G. Flötteröd, Capturing dependency among link boundaries in a stochastic dynamic network loading model, *Transp. Sci.* 49 (2) (2015) 420–431, <http://dx.doi.org/10.1287/trsc.2013.0504>.
- [50] K.W. Axhausen, A. Horni, K. Nagel, *The Multi-Agent Transport Simulation MATSim*, Ubiquity Press, 2016, <http://dx.doi.org/10.5334/baw>.
- [51] A. Agarwal, G. Lämmel, K. Nagel, Incorporating within link dynamics in an agent-based computationally faster and scalable queue model, *Transp. A* 14 (5–6) (2018) 520–541, <http://dx.doi.org/10.1080/23249935.2017.1364802>.
- [52] F. de Souza, O. Verbas, J. Auld, C.M. Tampère, A mesoscopic link-transmission-model able to track individual vehicles, *Simul. Model. Pr. Theory* 140 (2025) 103088, <http://dx.doi.org/10.1016/j.simpat.2025.103088>.
- [53] M. Mahut, M. Florian, *Traffic simulation with dynaflow*, 2010, pp. 323–361, http://dx.doi.org/10.1007/978-1-4419-6142-6_9.
- [54] S. Or, Mesoscopic and Hybrid Simulations in PTV Vissim, Technical Report, 2024, URL: <https://blog.ptvgroup.com/en/modelling-and-simulation-technology/mesoscopic-and-hybrid-simulations-in-ptv-vissim/>.
- [55] Aimsun, Aimsun Next User's Manual 22.0.1: Meso Traffic Modeling, Technical Report, 2024, URL: <https://docs.aimsun.com/next/22.0.1/UsersManual/MesoTrafficModelling.html>.
- [56] M. Mahut, *A Discrete Flow Model for Dynamic Network Loading* (Ph.D. thesis), Université de Montréal, 2001.

- [57] W.-L. Jin, Continuous formulations and analytical properties of the link transmission model, *Transp. Res. B* 74 (2015) 88–103, <http://dx.doi.org/10.1016/j.trb.2014.12.006>.
- [58] G. Mariotte, L. Leclercq, Flow exchanges in multi-reservoir systems with spillbacks, *Transp. Res. B* 122 (2019) 327–349, <http://dx.doi.org/10.1016/j.trb.2019.02.014>.
- [59] Y.-C. Ni, LIFT: Link-level interrupted flow traffic dynamics modeling, 2024, URL: <https://github.com/YCNI/LIFT.git>.
- [60] V.V. Gayah, C.F. Daganzo, Effects of turning maneuvers and route choice on a simple network, *Transp. Res. Rec.: J. Transp. Res. Board* 2249 (1) (2011) 15–19, <http://dx.doi.org/10.3141/2249-03>.
- [61] J. Espadaler-Clapés, E. Barmounakis, N. Geroliminis, Empirical investigation of lane usage, lane changing and lane choice phenomena in a multimodal urban arterial, *Transp. Res. A* 172 (2023) 103674, <http://dx.doi.org/10.1016/j.tra.2023.103674>.
- [62] E. Barmounakis, N. Geroliminis, On the new era of urban traffic monitoring with massive drone data: The pNEUMA large-scale field experiment, *Transp. Res. C* 111 (2020) 50–71, <http://dx.doi.org/10.1016/j.trc.2019.11.023>.
- [63] J. Espadaler-Clapés, E. Barmounakis, N. Geroliminis, Traffic congestion and noise emissions with detailed vehicle trajectories from UAVs, *Transp. Res. D* 121 (2023) 103822, <http://dx.doi.org/10.1016/j.trd.2023.103822>.